



Sustainable Scotland:

A Comparative Analysis of Development-Security

21st September 2026

ID: 2401049

Name: Joshua McManus

Institution: University of Buckingham

School: Faculty of Business, Humanities, and Social Sciences

BA(Hons) Security, Intelligence, and Cyber

Abstract

The modern international system treats sovereignty as the prerequisite for security. Scotland tests this assumption. It produces more renewable electricity than it consumes, possesses high-grade grazing land and vast fishing resources, and maintains net export capacity in major food categories. Its population nevertheless faces fuel poverty, food insecurity, and infrastructure vulnerabilities that resemble those of a small state more than a stable nation. This dissertation asks what Scotland's current Development-Security profile is in comparison to Norway and China, and how agency shapes such a profile.

To answer this question, the dissertation develops the PRESS framework, an analytical tool that assesses Development-Security across five interdependent dimensions: Populations, Resources, Ecosystems, Socio-economy, and Structures. The framework synthesises Realist and Critical Human Security scholarship through Giddens' structuration theory, reframing the referent object problem that has divided the field. It is applied comparatively across three policy domains: Energy Resource Management, Critical National Infrastructure, and Food Security.

The analysis finds that Scotland occupies a hybrid position. It resembles Norway in resource wealth and democratic values, but resembles China in supply chain dependence and the need for rapid state-coordinated action. Across all three domains, agency proves to be the binding constraint, preventing the operationalisation of Development-Security. Scotland can set ambitious climate targets but cannot control the market design required to deliver them. It can legislate on food policy but cannot regulate the trade and labour levers that determine food prices. It bears the human consequences of infrastructure failure without holding the regulatory authority to prevent it.

The finding is that agency shapes Development-Security, but not uniformly, and not as a simple function of sovereignty. Energy and Critical National Infrastructure demonstrate that agency is necessary for the capture of resource wealth and the control of risk. Food Security complicates this as while it is devolved, food insecurity persists, showing that formal authority alone does not deliver outcomes. Independence may remove the constitutional constraints identified in the empirical chapters, but its unlikely to resolve the domestic structural constraints of land ownership concentration, subsidy design, and rural poverty. EU accession, which Scotland's independence route would likely involve, would reconfigure this further: expanding Scottish agency over energy, strengthening it in critical infrastructure relative to the rest of the UK, and partially constraining it again over agriculture and fisheries. The dissertation concludes that independence would push Scotland toward the Norwegian model, but only with the institutional infrastructure that Norway built over decades. Independence without those institutions risks Chinese-style state dependency without Chinese state capacity.

Contents

I	Foundations	2
1	Introduction	3
2	Literature Review	5
3	Methodology	10
II	Main Body	15
4	Energy Resource Management	16
5	Critical National Infrastructure	20
6	Food Security	29
III	Analysis	37
7	The Independence Variable	38
8	Conclusion	41
IV	Reference List	42
	Reference List	43

Part I

Foundations

Introduction

The modern international system has been built on centuries of a singular premise: sovereignty is the price of statehood. Those without sovereignty, the logic goes, lack the capacity to secure themselves. Yet Scotland, a nation without sovereign control over its energy exports, grid stability, or supply chains, produces more renewable electricity than it consumes,¹ placing it among the world's green energy leaders. This directly challenges the sovereignty norm, raising the question of whether a non-sovereign entity can possess the resources of a state without the vulnerabilities of one. This dissertation argues that the answer is more nuanced than the norm allows. Scotland possesses the resources of a state but lacks the agency to convert them into security, and it is that gap (not sovereignty's absence as such) that produces its distinctive vulnerabilities. Scotland is a least-likely case for the claim that formal authority alone delivers Development-Security, and its failure to do so even here reveals what the sovereignty premise conceals. This dissertation examines how that absence shapes its Development-Security profile in ways the two sovereign comparators, Norway and China, cannot reveal.

Development-Security, a term often invoked but never holistically defined, has become a cornerstone of modern scholarship. From the security-development nexus, as theorised by Duffield² and Stern & Öjendal,³ to the UNDP's human security framework,⁴ academia has struggled to find a cohesive meaning that represents the securing and developing of both the state and the people which comprise it. The result is a scholarship divided by its referent object. Realists secure states whilst Critical Human and Security Scholars (what this dissertation will refer to outwith as CHSS) secure people - neither truly integrates both. This divide is not merely academic as it has practical consequences for territories like Scotland, which possess the resources of a state but lack the sovereignty to match.

As earlier stated, Development-Security is imagined by different readers in a variety of ways, which will often lead to conflicting understandings of the same work. Combining both sides of the literature produces the following working definition: *"The safeguarding of both the capacity and the capability for advancement, ensuring that progress does not hinder the ability of future generations to utilise the same mechanics of growth."* This definition is deliberately agency-neutral, because agency is the variable the dissertation tests. It is also multidimensional, because "capacity" and "capability" cannot be measured by any single indicator. PRESS is a tool created to analyse the impacts of policies across five dimensions: Populations, Resources, Ecosystems, Socio-economy, and Structures. The research question, therefore, is not whether security enables

¹The Scottish Government (2025) "*Energy Statistics for Scotland - Q3 2025*".

²Duffield, M. (2014) "*Global Governance and the New Wars: The Merging of Development and Security*".

³Stern, M. and J. Öjendal (2010) "'Mapping the Security-Development Nexus: Conflict, Complexity, Cacophony, Convergence?'"

⁴Jolly, R. and D. Ray (2006) "*The human security framework and national human development reports: A review of experiences and current debates*".

development, nor how development enables security, but how the distribution of agency shapes Scotland's Development-Security profile relative to Norway and China. This dissertation aims to:

- **Firstly**, define and operationalise the PRESS framework as an analytical tool for Development-Security, using structuration theory to reframe structure and agency;
- **Secondly**, map Scotland's current Development-Security profile across all five PRESS dimensions;
- **Thirdly**, analyse Norway and China as an axis of comparisons, using the same framework;
- **And finally**, test the hypothesis that independence would resolve the constitutional constraints identified in the empirical chapters, by analysing the Scottish Government's proposed institutional architecture against the Norwegian comparator.

The findings are intended to benefit Scottish Government policymakers designing independence transition plans, academic security studies seeking frameworks for small-state analysis, and comparative politics scholars examining green transitions in resource-rich sub-state entities. The dissertation argues that Scotland currently occupies a hybrid position. Closer to Norway in resource wealth and democratic values, but closer to China in supply chain dependence and need for rapid state-coordinated action. Independence would push Scotland toward the Norwegian model, but only with the institutional infrastructure (sovereign wealth fund, EU access, integrated CNI governance) that Norway built over decades. Independence without these institutions risks Chinese-style state dependency without Chinese state capacity. The structure-agency dynamic is central to the research as Scotland's constitutional constraints shape agency, while political mobilisation seeks to reshape those structures.

Literature Review

The biggest issue with Development-Security is the literature itself. As established, the scholarship at this nexus is fundamentally fractured. To understand why, and then establish the foundation upon which the PRESS framework rests, this review proceeds in five parts. First, it examines the foundational divide between Realist and CHSS schools, identifying the referent object problem within the field. Second, it interrogates the Development-Security Nexus, drawing on works previously mentioned to reveal the biopolitical tensions embedded within the merger of development and security. Third, it introduces Securitisation Theory⁵ and Energy Justice⁶ as essential framing mechanisms for understanding how issues become security threats and who benefits from transitions. Fourth, it grounds the analysis in Structuration Theory⁷ (as proposed by Giddens) demonstrating that structure and agency are co-constitutive, a resolution to the divide that informs the PRESS framework. Finally, it synthesises these threads into the PRESS framework itself, establishing it as a novel analytical tool for sub-state green transitions.

The Referent Object

The core debate within the literature can be summed up by understanding the idea of the 'Referent Object', that which is being Developed and Secured. Realist approaches, exemplified by Homer-Dixon, frame sustainability as strategic necessity. Homer-Dixon's analysis of environmental scarcity⁸ (supply-induced, demand-induced, and structural scarcity) established that resource competition can trigger conflict and state failure. His work provided a vocabulary for understanding how climate change and resource depletion translate into geopolitical instability. However, Homer-Dixon's framework assumes state sovereignty, rendering it ill-suited to sub-state entities like Scotland. Furthermore, realism's state-centrism limits analysis of how resource wealth translates to human security outcomes, a gap this dissertation addresses through the Populations and Socio-Economy dimensions of the PRESS Framework.

By contrast, CHSS Scholars, most notably Dalby⁹ and Duffield¹⁰ suggest that true security is rooted in the individual. Dalby's environmental security framework situates human wellbeing within planetary boundaries, arguing that climate change fundamentally challenges the state-centric security paradigm. Duffield, in his seminal work on the merging of development and security, demonstrated how global governance structures increasingly target populations rather than states, framing development as a security intervention. The modern human security frame-

⁵Buzan, B., O. Wæver, and J. De Wilde (1998) "*Security: A new framework for Analysis*".

⁶Jenkins, K. et al. (2016) "'Energy justice: A conceptual review'".

⁷Lippuner, R. and B. Werlen (2009) "'Structuration theory'".

⁸Homer-Dixon, T. F. (1999) "*Environment, scarcity, and violence*".

⁹Dalby, S. (2013) "*Security and environmental change*".

¹⁰Duffield, M. (2014) "*Global Governance and the New Wars: The Merging of Development and Security*" (cited on page 3).

work emerged from the 1994 UNDP Human Development Report,¹¹ which argued that security should shift from a 'narrow concern with territorial safety to a broader concern with the safety of people'. This foundational distinction has since shaped critical security studies and informs the UNDP's 2022 Special Report on Human Security,¹² which reaffirms that economic, food, health, and environmental insecurities are often more urgent than military threats for ordinary citizens. The central insight from this divide in research is not that one school is correct and the other incorrect, but that both capture essential but different dimensions of Development-Security. The state is not as black and white as researchers wish it could be; it requires secure resources and stable institutions, and people require secure livelihoods, food, and environmental conditions. The challenge, as this dissertation hopes to demonstrate, is analysing both simultaneously.

The referent object debate is not a disagreement about what security is, but a disagreement about what counts as security's failure condition. Realism treats the loss of territorial integrity as failure; CHSS treats the loss of individual welfare as failure. The dissertation's contribution is to demonstrate that for a sub-state entity, both failure conditions can be met simultaneously. A state can retain territorial integrity while systematically failing to secure its population's welfare. This is the condition PRESS is designed to detect.

The Security-Development Nexus

The Security-Development Nexus (SDN) is most famously associated with Duffield's analysis of the 'liberal peace',¹³ which shows how development interventions became securitised in the post-Cold War era, particularly in fragile states. Stern and Öjendal's later work¹⁴ demonstrates that the nexus operates differently across contexts. Sometimes development enables security. Sometimes security enables development. Sometimes the two are in tension. Venturi's analysis¹⁵ of the EU's involvement in the Sahel provides a contemporary illustration of that tension, showing that 'state security' interests frequently override 'human security' commitments. The PRESS framework treats this dynamic as a question of Structures, of which institutions hold the authority to securitise, and whose security commitments they serve.

The SDN has two limitations that matter for a sub-state case like Scotland. The first is its subject matter. The SDN was built to analyse fragile and post-conflict states. Its core concepts of liberal peacebuilding, securitised development, and state fragility presume a state that is weak, contested, or failing. Scotland is none of these. Its constitutional ambiguity creates vulnerabilities that resemble those of a small state, but it does not fit the SDN's diagnostic categories. The second limitation is more fundamental. The SDN orders its two terms in a way that carries conceptual consequences. In the phrase 'Security-Development', security comes first, and this ordering reflects a substantive priority. Across both its realist and its critical

¹¹UNDP (1994) "*Human development report 1994*".

¹²UNDP (2022) "*New threats to human security in the anthropocene demanding greater solidarity*".

¹³Duffield, M. (2014) "*Global Governance and the New Wars: The Merging of Development and Security*" (cited on page 3).

¹⁴Stern, M. and J. Öjendal (2010) "'Mapping the Security-Development Nexus: Conflict, Complexity, Cacophony, Convergence?'" (cited on page 3).

¹⁵Venturi, B. (2017) "*The Security-Migration-Development Nexus revised*".

variants, the SDN's analytical focus remains the security of the state against external threat, with development treated as an instrument or a consequence. Even where the SDN is critical of state-centrism, as in Duffield,¹⁶ the problem it diagnoses is still framed in terms of what security does to development. Development is the dependent variable.

This dissertation inverts the ordering. Development-Security takes development as the starting point and asks what follows for security. Sustainable development is not a downstream effect of security, but rather it may be classed as a direct condition of it. A state that cannot sustain its resource base, its institutions, or its population's livelihoods is not secure, regardless of its military capacity or its borders. Security is not the protection of the state from external threat but the durability of the conditions that make state and society possible. The two concepts are not sequential but in reality quite co-constitutive.

This distinction is the dissertation's central conceptual move. Where the SDN asks how security shapes development, Development-Security asks how development constitutes security. The referent object is the conditions under which the state and the individual both persist and it is counterintuitive to only analyse them as entities on their own. That is what the PRESS framework is designed to analyse, and it is why the framework cannot simply adopt the SDN's vocabulary. The SDN presumes the state-people separation that the next section examines. Development-Security treats the relationship between state and people as one of the structures it must assess.

Resolving the Structure-Agency Problem

The theoretical challenge at the heart of this dissertation is reconciling state-centric and human-centric perspectives. This dissertation suggests that it is not purely academic as it has practical consequences for how we analyse Scotland's Development-Security. Giddens' 1984 theory of structuration¹⁷ offers the conceptual resolution. Structuration rejects the notion that structures and agency are separate or oppositional. Instead, Giddens argues that structures are both the medium and the outcome of agency. Structures enable and constrain action, but action simultaneously reproduces and transforms structures. This is not a compromise between structure-centred (realist) and agency-centred (humanist) approaches rather instead, their synthesis as the two are ontologically inseparable, neither can exist without the other.

Structures do not cause agency, nor does agency cause structures. Rather, they constitute each other. Agency is always structured, and structures only exist through agency. This matters for Scotland because its constitutional status (a structure) constrains its agency on energy exports, grid governance, and agricultural policy. The Scottish Government cannot create a sovereign wealth fund, control energy pricing, or negotiate trade deals, as these are structural constraints that limit agency. But increased agency (which is the very motive for independence) could reshape those very structures. The relationship therefore is co-constitutive. Structures

¹⁶Duffield, M. (2014) "*Global Governance and the New Wars: The Merging of Development and Security*" (cited on page 3).

¹⁷Giddens, A. (1984) "*The constitution of society: Introduction of the theory of structuration*".

shape agency, but agency also shapes structures.

A distinction is needed here, because the dissertation uses 'agency' in two senses. The first is Giddens' ontological sense: agency as the universal human capacity to act within and upon structures. The second is operational: the specific capacity of a government to convert constitutional authority into Development-Security outcomes. This is the variable the dissertation tests empirically. Structuration explains why the two senses are related, a government's operational agency is itself structured by constitutional arrangements, but those arrangements are reproduced or transformed through political action. The framework the dissertation builds rests on this relationship. The five PRESS dimensions map onto the structure-agency dynamic. Structures (legal frameworks, institutional transparency) represent the structural pole. Populations and Socio-economy capture the outcomes of operational agency: Who benefits, Who bears costs, What capacities are developed. Resources and Ecosystems represent the material conditions within which the structure-agency dynamic plays out. None of these dimensions can be analysed in isolation; they are interdependent, and their interaction is precisely what Development-Security analysis must capture.

Securitisation Theory and Energy Justice

To understand how issues become security threats, this dissertation draws on the Copenhagen School's Securitisation Theory.¹⁸ Securitisation is not a property of issues themselves but a speech act. When an actor frames an issue as an existential threat requiring extraordinary measures, and the audience accepts this framing, the issue becomes securitised. This framework is essential for understanding how Scotland's energy dependency, grid vulnerability, and food import reliance might be constructed as security threats by policymakers, or how independence proponents might securitise constitutional constraint itself. The Structures dimension of PRESS operationalises this by examining legislative frameworks and institutional capacity to securitise issues.

Energy Justice provides a complementary lens, but it operates on a different level. Securitisation Theory explains how issues become security problems: the politics of framing. Energy Justice explains who bears the consequences of that framing: the politics of distribution. Jenkins¹⁹ argues that energy systems must be evaluated by distributive justice (who benefits), procedural justice (who decides), and recognition justice (whose voices are heard). This maps directly onto PRESS's Populations (who benefits and bears costs), Structures (who decides), and Socio-economy (community impacts). Securitisation tells us which issues get treated as threats; Energy Justice tells us whose interests are served when they do. This is particularly relevant to Scotland, where the transition from fossil fuels to renewables raises questions about diversifying 100,000 highly skilled North Sea oil and gas workers,²⁰ wealth distribution, and community voice, all questions independence would reshape.

¹⁸Buzan, B., O. Wæver, and J. De Wilde (1998) "*Security: A new framework for Analysis*" (cited on page 5).

¹⁹Jenkins, K. et al. (2016) "'Energy justice: A conceptual review'" (cited on page 5).

²⁰Robert Gordon University (2021) "*Majority of UK offshore workforce to be delivering low carbon energy By 2030*".

The PRESS Framework

This review has established that the Development-Security literature is divided by referent object, that the Security-Development Nexus provides a vocabulary for understanding the merging of these fields but has limits for sub-state analysis, that Structuration Theory reframes the structure-agency divide, and that Securitisation Theory and Energy Justice provide essential framing mechanisms. The PRESS framework synthesises these threads into five analytical dimensions. Each dimension draws on distinct theoretical traditions. Populations draws on realist concerns with demographic resilience. Resources and Ecosystems are analysed through a mixed lens, recognising that environmental conditions are simultaneously material constraints and social constructs. Socio-economy draws on liberal theory's concern with human welfare. Structures draws on critical security studies' attention to institutions and power relations. These are not theoretical silos but analytical entry points. Any empirical chapter applies all five dimensions because Development-Security is inherently multi-dimensional. The theoretical diversity reflects the complexity of the subject, no single framework captures the interplay of state capacity, human welfare, environmental limits, and institutional design. The PRESS framework is thus not a theoretical departure from the literature but its synthesis. Having established these theoretical foundations, the following Methodology chapter operationalises these insights into a comparative case study design, applying PRESS to Scotland, Norway, and China across Energy Resource Management, Critical National Infrastructure, and Food Security.

Methodology

Research Design

This dissertation employs a qualitative comparative case study design. The research question asks how the distribution of agency shapes Scotland's Development-Security profile in comparison to Norway and China. A qualitative approach is appropriate because the question concerns the capacity of governments to act, and that capacity cannot be captured by variables that treat constitutional authority and institutional capability as interchangeable. The comparative element is essential because it allows the PRESS framework to be tested across three constitutional arrangements: a devolved government with partial agency, a sovereign state with full agency, and a sovereign state with full agency under a different regime type. If the framework reveals systematic differences between these configurations, it demonstrates that Development-Security varies with agency rather than with sovereignty alone.

Case Selection

Three cases were selected to combine most-similar and most-different logics. Scotland is the primary case: a resource-rich sub-state entity with devolved powers but without sovereign control over energy markets, trade, or monetary policy. Norway serves as the primary comparator. A small, resource-rich, democratic nation with full sovereignty, a sovereign wealth fund, and EU adjacency. Scotland and Norway share similar population sizes, latitudes, and resource endowments, making them a most-similar pair in which the principal difference is constitutional status.

China serves as the second comparator. Scotland and China share an emerging-growth trajectory in renewable energy and a dependence on complex supply chains, but differ on regime type and on the constitutional status of the entity. Norway and China share full sovereignty but differ on regime type and growth stage. The three cases therefore form a triangle in which each pair shares one trait and differs on the others. Norway and Scotland share liberal-democratic institutions but differ on sovereignty; Scotland and China share an emerging-growth trajectory but differ on regime type; Norway and China share full sovereignty but differ on regime type and growth stage. This design allows agency to be isolated as the explanatory variable, because sovereignty, regime type, and growth stage each vary while the others are held constant in at least one pairing.

Because sovereignty co-varies with a range of institutional capacities (fiscal autonomy, EU access, and sovereign wealth governance), the Scotland-Norway comparison cannot isolate sovereignty as a single variable. It isolates the sovereignty package, the constitutional and institutional capacities that together distinguish a sovereign state from a devolved entity. That package, rather than sovereignty in the abstract, is one of the dissertation's explanatory variables. The other

is agency, defined in the following subsection. The relationship between the two is what the empirical chapters examine.

Agency

This dissertation defines agency as the capacity of a government to convert constitutional authority into Development-Security outcomes. Agency is not identical to formal legal competence. A government may hold formal authority over a policy domain yet lack the fiscal, institutional, or market levers to exercise it. Nor is agency identical to sovereignty. This distinction matters for Scotland, which holds devolved authority over several domains while lacking the fiscal and regulatory instruments that would make that authority effective.

Agency is operationalised through three indicators. The first is constitutional competence: whether the policy domain is reserved to Westminster, devolved to Holyrood, or shared. The second is fiscal autonomy: whether the government can raise and direct revenue independently within the domain, or depends on transfers and reserved taxation. The third is regulatory capability: whether the government controls the standards, tariffs, market design, and licensing regimes that determine outcomes within the domain. A government possesses high agency where all three are present, partial agency where constitutional competence is present but fiscal or regulatory control is not, and low agency where all three are absent. Scotland is expected to display partial agency across the three empirical domains; Norway and China are expected to display high agency, though through different institutional forms. The empirical chapters test these expectations.

Limits of Causal Inference

The dissertation's central claim is that agency shapes Development-Security outcomes. It does not claim that agency alone determines them, and the comparative design cannot establish that it does. Three cases cannot isolate a single causal variable from the range of factors that plausibly shape Development-Security outcomes. Norway's stronger performance relative to Scotland in several domains, for example, is consistent with the agency hypothesis, but it is also consistent with explanations grounded in fiscal capacity, institutional maturity, historical state-building, geographic position, and the terms on which Norway negotiated its relationship with the European Union. The dissertation does not attempt to rule these explanations out. It attempts to show that agency is the variable that best accounts for the specific pattern of outcomes observed across the three empirical domains: not that Norway and China outperform Scotland, but that the pattern of where Scotland performs well and where it performs poorly tracks the distribution of agency rather than other candidate variables.

This distinction matters for how the findings should be read. The dissertation treats its conclusions as analytical generalisations rather than statistical ones. Where the evidence supports a causal claim directly, the analysis states one. Where the evidence supports a mechanism or a pattern consistent with the agency hypothesis but does not exclude alternative explanations, the analysis says so. Where the evidence complicates the hypothesis, as the Food Security chapter

demonstrates, the complication is treated as a finding rather than a counter-example to be explained away. The comparative method is designed to test the agency hypothesis against three differently configured states and sub-state entities, not to establish that agency is the only variable that matters. The test is whether the three-case pattern is more consistent with the agency hypothesis than with the alternatives, and the analysis proceeds on that basis throughout.

Operationalisation of PRESS

The PRESS framework uses five dimensions because each captures a distinct class of failure. A country is secure when its population, its resource base, its ecosystems, its socio-economic distribution, and its constitutional structures are each sustainable. It is insecure when any one of them breaks down. These are distinct classes of failure, and none is reducible to another. A resource-rich country can be food-insecure. A democratic country can be ecologically unsustainable. An egalitarian country can lack the constitutional authority to act on its own behalf. A framework that omitted any one of the five dimensions would be blind to a class of failure the others cannot detect.

Each dimension is operationalised through specific indicators to ensure consistency across cases. **Populations** captures demographic resilience and human welfare through fuel poverty rates, food poverty, health outcomes, dependency ratios, and the distributional impacts of policy. This dimension asks who benefits and who bears costs. **Resources** captures material endowments and their governance through energy mix and generation capacity, domestic food production and import dependency, fossil fuel reserves, and supply chain vulnerabilities. This dimension asks what the country possesses, and how securely those resources are controlled. **Ecosystems** captures environmental conditions and sustainability through greenhouse gas emissions and carbon budgets, biodiversity indicators, land use and soil health, and the ecological impacts of energy and infrastructure projects. This dimension asks whether development trajectories are environmentally sustainable. **Socio-economy** captures economic structure and distribution through GDP per capita and diversification, inequality and poverty rates, employment in key sectors, supply chain retention, and fiscal mechanisms including taxation and sovereign wealth funds. This dimension asks whether development generates shared prosperity or concentrated wealth. **Structures** captures institutional and legal frameworks through constitutional and legislative competence, regulatory authority, the division of powers between devolved, national, and supranational bodies, planning regimes, and the capacity to securitise issues through legislation. This dimension asks who decides, and through what mechanisms.

Within PRESS, agency is assessed primarily through the Structures dimension, which captures constitutional competence, regulatory authority, and the division of powers. Socio-economy captures the consequences of agency for populations: who benefits, who bears costs, and whether the distribution of development reflects the exercise of agency or its absence. Populations, Resources, and Ecosystems capture the material and human conditions on which agency operates. Structures therefore names the capacity; the other dimensions register its effects. This mapping is why PRESS cannot be reduced to a list of factors. The dimensions are organised around the

relationship between the capacity to act and the outcomes that capacity produces, and that relationship is what the empirical chapters test.

Comparative Method

Each PRESS dimension is assessed across all three cases and then compared along two axes. The first is constitutional: does the dimension vary systematically with sovereignty? The second is agency-specific: does the dimension vary systematically with the capacity to convert authority into outcomes? Where the two axes diverge, that is, where a sovereign state and a sub-state entity differ on agency but not on outcomes, or vice versa, the agency hypothesis is complicated. Where they converge, the hypothesis is supported. The independence analysis then tests whether divergence is attributable to the constitutional dimension specifically, by asking what would change if Scotland's agency were expanded.

This comparative method also addresses the risk of what might be called descriptive comparison, the production of parallel country profiles that do not test a hypothesis. Each PRESS dimension is not simply described across the three cases; it is assessed against the expectation that high agency produces capability and that partial agency impairs it. Where a case departs from that expectation, the departure is treated as a finding rather than a nuisance, and the analysis asks what else is doing explanatory work.

Data Sources and Analysis

This study relies exclusively on secondary data. Data are drawn from three main categories. Official statistics include Scottish Government publications, UK Government DESNZ and ONS data, Statistics Norway, the Norwegian Ministry of Petroleum and Energy, and Chinese Government publications including the National Bureau of Statistics and the National Development and Reform Commission. International datasets include the International Energy Agency, World Bank, and United Nations Human Development Reports. Academic literature forms the second category, drawing on peer-reviewed journals and monographs across security studies, environmental governance, and comparative politics. Policy documents form the third category, including the Scottish Government's Building a New Scotland series, UK Energy Security Plans, Norwegian white papers, and Chinese Five-Year Plans.

The Building a New Scotland papers are treated as policy proposals rather than authoritative statements of fact. They are analysed for the institutional architecture they propose, and that architecture is assessed against the Norwegian comparator rather than accepted as a description of what independence would deliver. This treatment allows the independence analysis to test the dissertation's central claim, that independence without the institutions Norway built over decades would produce capability gaps rather than capability gains.

The analysis proceeds in four stages. First, each case is analysed individually across the five PRESS dimensions for each policy domain. Second, within-case analysis is followed by cross-case comparison to identify patterns and differences. Third, the independence variable is isolated by

asking which aspects of Scotland's PRESS profile would change under independence, drawing on the Building a New Scotland papers. Fourth, findings are synthesised to answer the core research question.

Limitations and Ethics

This study has several limitations. It relies almost exclusively on secondary data. Primary interviews would have enriched the analysis but were not feasible, due to the nature of the subject at hand. The comparative design is limited to three cases, which permits analytical generalisation about the relationship between agency and Development-Security but not statistical generalisation to all sub-state entities. The counterfactual analysis of independence is inherently speculative. To bound that speculation, it is anchored to the institutional architecture proposed in the Building a New Scotland papers rather than constructed from first principles. As no primary data was collected, nor was any subject interviewed, ethical approval was not required. The study involves no human participants, primary data collection, or sensitive personal data. All data are drawn from publicly available secondary sources.

Part II

Main Body

Energy Resource Management

Energy Resource Management is the clearest test of the agency hypothesis, because no other domain so sharply separates what a state can produce from what it can control. Scotland generates more renewable electricity than it consumes and cannot protect its households from fuel poverty. The gap between those two facts is not an engineering problem. Rather, it is a constitutional one, and it is the central finding of this chapter.

Population

An energy system that generates high aggregate output while imposing severe financial burdens on domestic consumers represents a developmental failure. Scotland demonstrates the failure; Norway and China demonstrate its remedy. In 2024, one third of Scottish households lived in fuel poverty.²¹ That is 732,000 households, half of whom endured extreme fuel poverty, with a median fuel poverty gap of £1,030.²² Rural and island communities face the sharpest burden, where the Scottish Minimum Income Standard applies cost uplifts ranging from 8.4% to 32.8% to reflect harsh climate exposure and older housing stock.²³ The mechanism behind this paradox is not physical. Retail electricity pricing remains bound to marginal fossil gas rates under current market design, forcing Scottish consumers to pay premium tariffs for power generated in their own backyard.²⁴ Physical generation is not the problem, market architecture is, and market architecture is not devolved.

Norway and China avoid this outcome by controlling the mechanism that produces it. Norway insulates households through direct electricity bill subsidies funded from surplus petroleum revenues during European market shocks, sheltering its citizens from external price volatility,²⁵ and its hydroelectric base, which supplies over 80% of national production,²⁶ keeps domestic supply decoupled from fossil fuel markets. China manages population energy security through centralised state regulation. Retail tariffs for urban households remain below cost via state-administered cross-subsidisation from industrial users.²⁷ Both sovereign states possess the fiscal and regulatory instruments that convert energy wealth into household welfare. Scotland possesses this wealth but lacks the instruments to materialise it.

²¹Energy Action Scotland (2024) “*Fuel poverty*”.

²²Thompson, Z., E. Iordanidou, and A. Scrimgeour (2026) “*Scottish House condition survey 2024: Key findings*”.

²³Padley, M., A. Bryan, et al. (2026) “*The cost of remoteness: Reflecting higher living costs in remote rural Scotland when measuring fuel poverty 2024 update*”.

²⁴Barnes, Bräuer, and Tahir (2024) “*GB wholesale electricity market reform: Impacts and opportunities for Scotland*”.

²⁵Government, N. (2025) “*Revidert nasjonalbudsjett: Norgespris på strøøm og fjernvarme*”.

²⁶International Energy Agency (2026) “*Norway - countries & regions*”.

²⁷Liu, S., W. Ye, et al. (2026) ““Evaluating and optimizing residential electricity price tiers considering income redistribution equity under cross-subsidies mechanisms””.

Resources

The critical difference in resource security is not abundance but dispatchability. Scotland's renewable endowment is world-class and structurally intermittent. Norway's is world-class and dispatchable and China's is mixed and held in reserve by state design. The three cases reveal that dispatchability is produced by the institutional capacity to plan generation portfolios across time. Scotland possesses exceptional renewable natural capital in onshore and offshore wind, maritime hydrodynamics, and hydro-storage capacity, and by 2024 low-carbon sources accounted for the vast majority of electricity generated in Scotland.²⁸ The Scottish strategy targets 20 GW of onshore wind by 2030 and up to 40 GW of operational offshore wind by 2040 through the ScotWind leasing pipeline,²⁹ supported by legacy pumped hydroelectric storage such as Cruachan Power Station³⁰ and a green hydrogen strategy targeting 25 GW by 2045.³¹

These targets rest on a structural weakness. Scotland's grid depends on wind and solar assets that cannot be dispatched on demand, and the institutional capacity to compensate for that intermittency across the wider system is not devolved. Norway avoids this problem through a dual-vector system where domestic power generation derives from flexible hydropower, paired with extensive offshore hydrocarbon reserves on the Norwegian Continental Shelf. The dispatchability of Norway's vast hydro reservoirs provides built-in system flexibility.³² China takes a different route which can be seen through the 15th Five-Year Plan which codifies a dual-track strategy that scales solar and wind manufacturing while maintaining coal-fired generation as a dispatchable baseload reserve.³³ Both sovereign states treat energy security as a strategic obligation that their institutions are equipped to plan for while Scotland still treats it as a market outcome.

Ecosystems

The ecosystems dimension exposes a pattern that is not unique to Scotland, as all three states face a gap between environmental commitment and environmental control. The gap takes a different form in each case, and the differences reveal what agency can and cannot resolve. Scotland's gap is constitutional. Its statutory mandate to achieve Net Zero greenhouse gas emissions by 2045 implies an aggressive 2036-2040 budget reduction of 80.0% below the 1990 baseline, a 59.0% contraction from 2023 levels.³⁴ The trajectory depends on transport, heating, and agricultural decarbonisation alongside Negative Emissions Technologies whose regulation is constitutionally reserved to the British Government.³⁵ Scotland has set the target but once again does not control the levers required to deliver it.

²⁸The Scottish Government (2025) "*Energy Statistics for Scotland - Q3 2025*" (cited on page 3).

²⁹Scottish Renewables (2023) "*Onshore wind sector deal for Scotland*".

³⁰Drax Group (2026) "*Pumped Storage Hydro*".

³¹Scottish Government (2022) "*Hydrogen action plan*".

³²Arvesen, A. et al. (2025) "'On the potential role of flexible Norwegian hydropower in managing challenging renewable energy variability events in Europe'".

³³National Development and Reform Commission (2026) "*新型能源体系建设‘十五五’规划*".

³⁴Scottish Government (2023) "*Scottish Greenhouse Gas Statistics 2023*".

³⁵Climate Change Committee (2025) "*Progress in reducing emissions in Scotland 2025 report to Parliament*".

Norway's gap is externalising. The state enforces rigorous domestic environmental mandates and municipal transport electrification, yet its dependency on global hydrocarbon exports effectively externalises its downstream combustion emissions. This friction is further complicated by the October 2021 Norwegian Supreme Court ruling that the Fosen onshore wind farm violated indigenous Sámi rights under Article 27 of the UN ICCPR^{36,37} a case that shows even full sovereign agency does not resolve the distributive conflicts that energy transitions produce. China's gap is contradictory. It navigates ecological security through a dual-target paradigm, aiming to peak emissions before 2030 and attain neutrality by 2060,³⁸ supplemented by localised "Ecological Protection Red Lines," yet its simultaneous expansion of coal infrastructure underscores a persistent tension between localised environmental pressure and national climate targets. In each case, the gap between commitment and control is real but the form in which it manifests differs.

Socio-Economy

The socio-economic test of the agency hypothesis is whether a state can capture the value its own resources generate. Scotland's Green Industrial Strategy, launched in September 2024, targets five core areas: wind economy, clean hydrogen, CCUS, green financial services, and energy-intensive manufacturing.³⁹ The strategy has not translated into domestic economic security. An estimated £2.5 billion to £4.5 billion in capital investment is required for port upgrades against public allocations under £600 million.⁴⁰ The funding gap is structural, and it is compounded by a constitutional one. Scotland cannot impose local content requirements because offshore licensing and market design are reserved to Westminster.

Norway and China capture the value of their energy resources because they possess the fiscal and industrial authority to do so. Norway's fiscal regime combines an ordinary corporate tax rate of 22% with a special petroleum tax rate of 56%, yielding a 78% marginal rate on upstream profits, while allowing immediate CapEx write-offs that mean the state effectively funds 78% of upfront project costs.⁴¹ These revenues feed the Government Pension Fund Global, which holds over £1.75 trillion in assets and transforms exhaustible hydrocarbons into permanent capital for future generations.⁴² China builds socio-economic security through clean-energy industrial value chains and achieving dominant global market share in solar PV fabrication, wind turbine manufacturing, battery chemistry processing, and rare earth refining through State-Owned Enterprises.⁴³ Unlike Scotland, both comparator states possess the authority to convert resources into economic security.

³⁶Norges Domstoler (2021) "*Supreme Court of Norway, Statnett SF et al. v. Sør-Fosen sjeite*".

³⁷OHCHR (1985) "'International Covenant on civil and political rights'".

³⁸Ministry of Ecology and Environment (PRC) (2019) "国务院新闻办公室发布《碳达峰碳中和的中国行动》白皮书 (附全文)".

³⁹Scottish Government (2024) "*Green industrial strategy*".

⁴⁰Transition Economics (2024) "*Scotwind: The investment needed to secure manufacturing jobs in Scotland*".

⁴¹PwC (2026) "*Norway - Petroleum tax regime*".

⁴²Norges Bank Investment Management (2019) "*Oljefondet*".

⁴³Global Energy Monitor (2024) "*China continues to lead the world in wind and solar, with twice as much capacity under construction as the rest of the world combined*".

Structures

The Structures dimension is where the agency argument becomes constitutionally concrete. Scotland's energy governance operates within an asymmetric framework created by the Scotland Act devolution settlement.⁴⁴ Devolved competencies include spatial planning, environmental consenting, building standards, heat policy, land-use management, and economic development agencies. Reserved competencies include macro-electricity market design, wholesale pricing, transmission tariff regulation, offshore licensing regimes, and carbon market structures. Scotland can set ambitious climate targets and issue planning consent, but it cannot control the market design and grid transmission structures required to deliver them. Strategic spatial initiatives such as the joint National Energy Systems Operator Strategic Spatial Energy Plan and Regional Energy Strategic Plans attempt to bridge this gap. However, these organisations cannot act because they are coordination mechanisms.

Norway and China achieve coherence through unified legal authority. Norway operates a streamlined, unitary governance structure centred in the Ministry of Petroleum and Energy. Parliament can rapidly update tax codes and align exploration licensing, grid planning, and sovereign fund management within a single jurisdictional framework. China operates under a hyper-centralised governance framework anchored by the National Development and Reform Commission, which allows the state to fast-track major infrastructure approvals, build ultra-high-voltage transmission networks, and deploy domestic industrial capital without multi-jurisdictional legal friction.⁴⁵ Both sovereign states possess what Scotland lacks, a single point of authority where energy policy is designed, financed, and delivered.

Conclusion

The comparative evidence establishes one finding. Agency shapes whether resource wealth becomes energy security. Norway and China convert energy abundance into household welfare, industrial capacity, and strategic reserves because they possess the constitutional, fiscal, and regulatory instruments required to do so. Scotland produces more renewable electricity than it consumes and cannot convert that abundance into population welfare. Thus, the reason is constitutional. Agency over energy is distributed across two levels of government without a mechanism for integration, and the level that holds the resources does not hold the levers.

The domain-specific finding is that energy is the least ambiguous case for the agency hypothesis. Unlike food security, where constitutional authority and policy outcomes diverge, energy shows a direct correspondence between the location of agency and the location of capability. Where Scotland holds agency (planning, consenting, target-setting), it exercises it competently. Where it lacks agency (market design, fiscal transfer, offshore licensing), capability fails. Norway and China confirm the pattern by contrast. Each holds agency in a unified form, and each converts resources into security outcomes.

⁴⁴British Government (1998) "*Scotland Act 1998*".

⁴⁵中华人民共和国国家发展和改革委员会 (2025) "关于促进电网高质量发展的指导意见(发改能源〔2025〕1710号)" -国家发展和改革委员会".

Critical National Infrastructure

Critical National Infrastructure is the physical substrate of Development-Security. It connects economic capacity, ecological resilience, and human welfare. When essential infrastructure fails, the consequences propagate through populations, economies, and institutions with a speed that governance structures cannot match. Scotland faces a specific version of this problem. The physical networks that sustain its population remain reserved to Westminster, while statutory responsibility for their operational resilience sits with Holyrood. This disjunction between authority and dependency produces a governance gap that neither entity can close unilaterally. The consequence is a structural vulnerability in Scotland's Development-Security profile that no amount of capital investment can resolve.

The chapter's finding is that coherence, meaning the alignment of legal authority with operational responsibility, is a component of agency, and that agency is the key factor in CNI resilience. Norway and China demonstrate this by contrast with Scotland. But coherence alone is not sufficient, even unified sovereign authority can produce outcomes that violate a state's own commitments, as the Fosen case demonstrates. The argument is therefore not that coherence guarantees resilience, but that its absence makes resilience structurally precarious regardless of what else is in place.

Populations

The Populations dimension tests the consequences of the coherence gap at its sharpest point: physical isolation. When essential infrastructure fails, the resulting disruption becomes human insecurity, and the effect is sharpest among isolated rural and island populations where alternatives do not exist. Two distinct physical dependencies expose the same governance gap. Subsea telecommunications cables and strategic road corridors. In both cases, Scotland bears the human consequences of failure without holding the authority to prevent it.

Scotland's island territories demonstrate this precisely. Communities in the Outer Hebrides and Northern Isles depend entirely on subsea telecommunications cables for broadband, cellular coverage, landline communication, and emergency service dispatch. When those cables are severed, the consequences are immediate. Storm Amy⁴⁶ made this clear in late 2025. This proves that digital connectivity is critical national infrastructure, and not some commercial amenity or luxury as the severance of a single subsea cable disables:

- Emergency services (999 call routing), impairing dispatch mechanisms for ambulance, coastguard, and fire services.
- Primary healthcare facilities and local NHS clinics, which lose access to central medical

⁴⁶Pizzuto-Pomaco, J. (2025) ““Internet cut off on Tiree after Storm Amy damages cable””.

databases, electronic prescription systems, and remote diagnostics.

- Commercial life, as point-of-sale banking terminals, ATMs, and local business networks collapse.
- Critical civic links, including ferry booking portals, agricultural supply chains, and school digital systems.

A parallel physical bottleneck exists on the A83 trunk road at the Rest and Be Thankful in Glen Croe. The A83 is one of only two strategic east-west transport corridors connecting Argyll and Bute to the Central Belt via the A82 junction at Tarbet. Alternative northern routes require detours of up to 30 miles. The A83 functions as the primary regional logistics pipeline. Commercial logistics make up nearly 30% of its total daily volume, carrying nearly 500 heavy goods vehicles and 800 light goods vehicles every day.⁴⁷ It captures the overwhelming majority of heavy freight moving between the Central Belt and the region's southern and western hubs, and it is essential for fresh seafood, heavy timber, and distilling exports.⁴⁸ The steep slopes of Beinn Luibhean above the A83 suffer from chronic hillside instability. Major landslides, most severely in August and September 2020,⁴⁹ have repeatedly buried the carriageway under thousands of tonnes of rock and mud.

Transport Scotland maintains access during closures via the Old Military Road, a single-track local diversion along the glen floor.⁵⁰ It is also advancing a £500 million permanent Long-Term Solution. The preferred design, the “Brown Option,” involves a 1.4 km debris flow shelter (a concrete canopy over the carriageway backed by a reinforced protection wall and catch pit to deflect falling landslip material safely into the valley⁵¹). After 14 years of design iterations, public local inquiries, and statutory assessments under the Design Manual for Roads and Bridges, main construction remains uninitiated. The delay forces regional businesses and hauliers onto long detours through Glen Kinglas or along local roads prone to crumbling under heavy traffic. The cumulative losses to the Argyll economy run into the millions.⁵² The finding is not that Transport Scotland lacks competence however the decision architecture within which Transport Scotland operates disperses authority across statutory assessments, public inquiries, and funding approvals in ways that delay action even when the operational need is unambiguous. Scotland has authority over trunk roads but naturally doesn't have authority over the pace at which that authority can be exercised.

Norway and China both treat infrastructure resilience as a state obligation, though they approach execution and governance differently. Norway mandates public safety preparedness through statutory municipal resilience standards and civil protection requirements, heavily investing in physical redundancy by replacing vulnerable ferry routes with subsea road tunnels like

⁴⁷Transport Scotland (2017) “*Record of Determination*”.

⁴⁸Dyet, G. et al. (2022) “*A83 rest and be thankful story map*”.

⁴⁹BBC News (2020) ““Significant’ landslide forces closure of rest and be thankful””.

⁵⁰Transport Scotland (2023) “*Options assessment report - January 2023 - A83 Access to Argyll and Bute - Introduction*”.

⁵¹Transport Scotland (2023) “*Access to Argyll and Bute (A83)*”.

⁵²Sunday Post (2018) “*Argyll traders call for solution to A83 closures that cost up to £60,000 a day*”.

the Ryfast project. Norway's civil protection framework enforces a risk-based resilience model. This strategy prioritises geographical route diversification and public-private partnerships with commercial operators to ensure continuous connectivity through alternative physical networks. China manages rural population vulnerability through state-enforced infrastructure standards. Under central directive, State Grid Corporation and state telecommunications operators must build redundant power and cellular links to remote agricultural counties, absorbing operating losses through state cross-subsidisation.⁵³ Both sovereign states possess a single point of authority where resilience is designed and funded. Scotland's island and rural communities depend on a coordination mechanism that no single authority owns.

Resources

The Resources dimension tests where coherence fails most visibly. Scotland's green energy trajectory depends on expanding its electricity transmission network creating a transmission capacity constraint. Renewable power generated in the north must reach industrial and urban demand centres in the south, and Great Britain's inland transmission capacity faces structural bottlenecks across the Anglo-Scottish border.⁵⁴ The response has been a suite of subsea High Voltage Direct Current corridors off the Scottish coast,⁵⁵ with combined capital costs exceeding £10 billion: the Western HVDC Link, Eastern Green Link 1, 2, and 4. The Western Link, operational since 2019, is a £1.2 billion, 2,250 MW link from Hunterston to Flintshire Bridge, and it has experienced prolonged technical outages resulting in a £158 million regulatory penalty against its joint operators National Grid and ScottishPower due to severe wind generation curtailment.⁵⁶ Eastern Green Link 1 is a £2.5 billion, 2 GW link between Torness and Hawthorn Pit, scheduled for completion in 2029.⁵⁷ Eastern Green Link 2 is a £4.3 billion, 2 GW link from Peterhead to Drax and the largest single electricity transmission project in British history.⁵⁸ Eastern Green Link 4 is a 640 km, 2 GW link between Fife and Norfolk, targeting operation by 2033.⁵⁹

These corridors are evidence of the absence of Scottish grid capability, because they exist to bypass a bottleneck that Scottish authorities cannot resolve unilaterally. Under existing constraints, the National Energy System Operator must routinely pay compensation fees to Scottish wind farm operators to turn off generation during periods of high wind. NESO estimates that operationalising links like EGL1 will reduce annual consumer constraint payments by over £870 million.⁶⁰ The figure measures, in the opposite direction, the cost of the coherence gap. The transmission network that would avoid those payments cannot be built without authority Scotland does not hold.

⁵³Cyberspace Administration of China (PRC) (2016) “中华人民共和国电信条例”.

⁵⁴NESO (2022) “*Scottish boundaries — national energy system operator*”.

⁵⁵OFGEM (2024) “*Proposed anglo-scottish electricity superhighway is first to clear final fast-track funding hurdle*”.

⁵⁶SP Energy Networks (2024) “*Western HVDC link*”.

⁵⁷Eastern Green Link 1 (2023) “*About EGL1*”.

⁵⁸Eastern Green Link 2 (2024) “*Go-ahead for UK's biggest subsea connection project*”.

⁵⁹Hakimian, R. (2026) ““£3Bn of contracts signed on Fourth England to Scotland electricity ‘superhighway’””.

⁶⁰Dunshea, J. (2024) “*Decision eastern green link 1 - project assessment*”.

In water utilities, Scottish Water manages a regional infrastructure portfolio that delivers over 1.5 billion litres of drinking water daily and treats domestic wastewater across the nation.⁶¹ Maintaining this network requires sustained capital allocations exceeding £600 million per investment cycle to replace legacy piping, protect drinking catchments from agricultural runoff, and ensure treatment facilities resist localised climate hazards.⁶²

Norway manages its infrastructure resources through a sovereign asset protection model. Statnett operates an integrated hydro-dominated grid linked to continental Europe via high-capacity subsea interconnectors,⁶³ and Norway does not suffer from internal grid constraint costs because it controls its own grid architecture. Subsea oil and gas pipelines operated by Gassco are monitored by state security services and naval patrols, ensuring physical integrity across offshore fields.⁶⁴ China secures its infrastructure resources through state-directed construction of Ultra-High Voltage direct-current transmission lines⁶⁵ spanning thousands of kilometres from inland renewable hubs in Xinjiang and Inner Mongolia to coastal megacities,⁶⁶ bypassing regional market fragmentation under direct state oversight. Both sovereign states design their infrastructure to serve strategic ends. Scotland's infrastructure is designed by a market and regulated by a state that is not its own.

Ecosystems

The Ecosystems dimension tests the limit of coherence. Infrastructure expansion must be reconciled with statutory environmental commitments, and the question the dimension asks is whether a state can enforce its own environmental standards across the infrastructure that most affects ecological outcomes. All three cases face a gap between environmental commitment and environmental control, and the Fosen ruling demonstrates that even full sovereign agency does not resolve the distributive conflicts that infrastructure transitions produce.

Scotland's gap is constitutional. Climate projections indicate increased frequency of extreme rainfall events and wetter winters,⁶⁷ accelerating soil saturation, increasing slope failure risk along transport corridors such as the A83,⁶⁸ and raising scour risks around bridge foundations.⁶⁹ The Scottish National Adaptation Plan and the Climate Ready Scotland framework require all public bodies and infrastructure operators to integrate climate adaptation into strategic risk management.⁷⁰ Expanding terrestrial and subsea CNI creates environmental friction that must be reconciled with statutory Net Zero and biodiversity mandates. Constructing overhead transmission lines, converter stations, and subsea cables requires compliance with Scotland's National

⁶¹Scottish Water (2026) *"Our vital role"*.

⁶²Scottish Water (2026) *"Strategic review of charges 2027-33"*.

⁶³Statnett (2025) *"Interconnectors"*.

⁶⁴Norwegian Petroleum Safety Authority (2026) *"Integrating safety and security"*.

⁶⁵People's Daily (2025) *"China puts 45 ultra-high-voltage projects into operation"*.

⁶⁶CGTN (2026) *"China's west-to-East power project delivers over 1 trillion kWh to Jiangsu"*.

⁶⁷Met Office (2019) *"UKCP18 Factsheet: Scotland"*.

⁶⁸Klaver, N. R. et al. (2024) *"Mass movement hazard & global climate change: Physical-deterministic modelling study of the Rest and be Thankful pass, Scotland UK"*.

⁶⁹Sayers, P. et al. (2021) *"Third UK Climate Change Risk Assessment Technical Report"*.

⁷⁰British Government (2009) *"Climate Change (Scotland) Act 2009, Section 53: Programmes for adaptation to climate change"*.

Planning Framework 4.⁷¹ In Scotland, consenting for large onshore electricity infrastructure is governed by the Electricity Act 1989, with the Scottish Ministers as decision-makers.⁷² These frameworks require major infrastructure projects to address biodiversity and environmental impacts through three mechanisms: reconnecting fragmented upland and woodland habitats via dedicated green corridors and biodiversity stepping stones;⁷³ minimising benthic disturbance and seabed disruption during subsea cable trenching and rock placement;⁷⁴ and mitigating landscape, visual, and acoustic impacts on protected scenic areas.⁷⁵ The framework is coherent. Its enforcement depends on reserved infrastructure whose expansion it cannot compel.

Norway's gap is institutional, and Fosen shows its limits. Norway balances ecological preservation and infrastructure deployment through statutory municipal environmental assessments under the Forskrift om konsekvensutredninger (Regulations on Impact Assessments).⁷⁶ The Fosen wind farm case tested this framework against indigenous Sámi reindeer herding rights under Article 27 of the UN ICCPR.⁷⁷ The Norwegian Supreme Court found that the licences violated those rights, leading the government to apologise and initiate remedial measures.⁷⁸ The ruling demonstrates both the strength and the limit of coherence. Norway's unified legal framework can address ecological and indigenous rights claims within a single constitutional order, which Scotland cannot do, because the relevant competencies are split across two governments. But unified authority did not prevent the violation in the first place, it only provided the mechanism for remedy after the fact. Coherence enables correction, it does not guarantee prevention.

China's gap is contradictory. It addresses infrastructure ecology through its "Ecological Protection Red Lines" policy, establishing strictly protected environmental zones across key river basins,⁷⁹ and it enforces the policy through the same centralised apparatus that drives infrastructure expansion. China's ecological governance and its infrastructure governance answer to the same authority. Scotland's do not. But the Chinese case also shows that unified authority can pursue contradictory ends simultaneously, because the red lines and the coal expansion are both products of the same centralised system. Coherence concentrates power but lacks the ability to create coherence of purpose.

Socio-Economy

The Socio-Economy dimension tests the other components of agency alongside coherence. Financial capacity and technical capability matter to CNI resilience as much as legal authority

⁷¹Scottish Government (2023) "*National Planning Framework 4*".

⁷²British Government (1989) "*Electricity Act 1989*".

⁷³Scottish Government (2025) "*Biodiversity: planning guidance*".

⁷⁴Scottish Government (2015) "*Scotland's National Marine Plan: A Single Framework for Managing Our Seas*".

⁷⁵Department for Energy Security and Net Zero (2026) "*National Policy Statement for Electricity Networks Infrastructure (EN-5)*".

⁷⁶Klima- og miljødepartementet and Kommunal- og distriktsdepartementet (2017) "*Forskrift om konsekvensutredninger*".

⁷⁷Norges institusjon for menneskerettigheter (2025) "*Fosen-aksjonistene vant frem i Høyesterett*".

⁷⁸Norges institusjon for menneskerettigheter (2024) "*Evalueringen av Fosen-saken*".

⁷⁹Chinese Academy of Sciences (2025) "*New Study Reveals China's Ecological Red Lines Offer Blueprint for Biodiversity Protection*".

does, and the two examples in this dimension show that Scotland's coherence gap compounds with gaps in fiscal and technical capacity rather than sitting alongside them.

Resilient CNI requires major capital expenditure balanced against consumer affordability and supply chain constraints. Scotland's electricity network operators have initiated massive capital programs to deliver Net Zero grid readiness. SP Energy Networks is deploying a £12 billion investment to overhaul central and southern Scotland's transmission grid.⁸⁰ The operator estimates this investment will contribute £2 billion directly to UK GDP,⁸¹ provide a £5.4 billion boost to regional supply chains,⁸² and support 1,400 direct engineering jobs.⁸³ By building subsea export cables to reduce wind curtailment payments, these projects are projected to reduce average household energy bills by £167 by 2030 following initial capital recovery periods.⁸⁴

These capital allocations face severe supply chain concentration and technical bottlenecks. Constructing specialised HVDC links such as EGL1-4 relies on a limited number of international high-voltage equipment manufacturers, primarily Siemens Energy and Prysmian.⁸⁵ Global supply chain constraints in specialised converter transformers, high-voltage subsea cables, and off-shore installation vessels have extended lead times and increased raw material costs, creating project delay risks.⁸⁶ Scotland cannot manufacture the components its grid requires. It cannot compel manufacturers to prioritise its projects. It can only bid for scarce capacity in a global market. The fiscal and industrial instruments that would let Norway or China shelter such projects from global market conditions are held by the UK Government, not by Scottish Ministers.

The convergence of physical infrastructure with digital control systems introduces acute cyber-physical vulnerabilities. Modern energy grids, water treatment plants, and transport networks rely on Industrial Control Systems and Supervisory Control and Data Acquisition software, often managed by third-party Managed Service Providers (MSPs).⁸⁷ The UK Cyber Security and Resilience Bill seeks to mandate strict risk assessments, incident reporting, and network protection duties across these supply chains.⁸⁸ As it is reserved legislation, it will apply to Scotland whether or not Scottish Ministers consider its provisions adequate.

The digital transition requires a highly skilled technical workforce capable of securing critical networks against ransomware, distributed denial-of-service attacks, and state-sponsored cyber

⁸⁰SP Energy Networks (2026) "*Our RII0-T3 Business Plan*".

⁸¹ScottishPower (2025) "*ScottishPower to invest up to a record £12bn in Scotland's electricity grid*".

⁸²ScottishPower (2025) "*ScottishPower to invest up to a record £12bn in Scotland's electricity grid*" (cited on page 25).

⁸³ScottishPower (2025) "*ScottishPower to invest up to a record £12bn in Scotland's electricity grid*" (cited on page 25).

⁸⁴World Ports Organization (2026) "'Green light from Ofgem for 2 GW subsea electricity superhighway'".

⁸⁵Imperial Consultants (2025) "*Dogger Bank Final Report: HVDC Supply Chain and Procurement*".

⁸⁶International Energy Agency (2025) "*Building the Future Transmission Grid: Strategies to Navigate Supply Chain Challenges*".

⁸⁷UK Tech News (2026) "'Over three-quarters of CNI organisations report repeated supply chain compromise'".

⁸⁸UK Government (2025) "*Cyber Security and Resilience (Network and Information Systems) Bill: Summary*".

intrusions. Scotland benefits from academic centres of excellence in cyber security,⁸⁹ but operates within a global market facing severe technical skill shortages. China's technical data reveals that by 2027, its national cybersecurity talent gap will reach 3.27 million professionals against an annual university graduation rate of only 30,000 specialists.⁹⁰ The comparison is instructive in an unexpected direction. Even a state with full sovereign agency over education, industrial policy, and state enterprises faces a structural workforce gap in the cyber domain. Full agency relocates capability risk but is unable to fully eliminate it.

Norway manages these socio-economic dynamics through sovereign financial reserves. The Government Pension Fund Global holds over \$2.2 trillion USD in assets.⁹¹ The Norwegian state can absorb inflation shocks in material costs and fund specialised maritime cyber defence units without overburdening domestic utility ratepayers. China secures its infrastructure socio-economy through state-owned enterprise control. By maintaining state ownership over cable manufacturing, steel production, grid construction, and digital hardware,⁹² China eliminates foreign supply chain leakage and locks capital expenditures within its domestic economy. Both sovereign states deploy fiscal instruments to protect their infrastructure programmes from external shocks. Scotland cannot, because the fiscal levers that would enable such protection remain reserved to Westminster. The coherence gap compounds with fiscal and technical gaps, and that the three reinforce each other rather than operating independently.

Structures

The Structures dimension reveals the governance gap in its purest form. Where the other four dimensions show Scotland experiencing the consequences of dependency, Structures shows the constitutional architecture that produces them. CNI governance in Scotland is defined by an asymmetric statutory division created by the Scotland Act devolution settlement. Infrastructure oversight splits between reserved competencies held by the UK Government at Westminster and devolved competencies managed by the Scottish Government at Holyrood.⁹³ Reserved CNI sectors include Energy, Telecommunications, Civil Nuclear, Defence, and broad National Security Strategy. Key regulatory and technical authorities operating within these reserved domains include the Department for Energy Security and Net Zero, the Office of Gas and Electricity Markets, the Office of Communications, the National Cyber Security Centre,⁹⁴ the National Protective Security Authority,⁹⁵ and the Cabinet Office COBR Directorate.⁹⁶ Devolved CNI sectors encompass Water, Healthcare, Food Supply, Spatial Planning, Local Emergency Response, and Trunk Roads.

This split produces operational friction across critical sectors. The Scottish Government

⁸⁹Edinburgh Napier (2024) "*Edinburgh Napier recognised as an academic centre of excellence in cyber security education*".

⁹⁰CGTN (2022) "*China's demand for cybersecurity talents to reach 3.27 million By 2027*".

⁹¹Norges Bank Investment Management (2026) "*Half-year report 2026: Government Pension Fund Global*".

⁹²新浪财经 (2026) "*“国网甘肃物资事业部多措并举护航重点项目建设”*".

⁹³UK Government (1998) "*Scotland Act 1998, Schedule 5: Reserved Matters*".

⁹⁴National Cyber Security Centre (2025) "*The NCSC at a glance*".

⁹⁵National Protective Security Authority (2026) "*Critical National Infrastructure*".

⁹⁶UK Government (2026) "*Written questions and answers: COBR Directorate responsibilities*".

maintains statutory responsibility for the operational resilience of devolved public services such as NHS Scotland’s 14 territorial Health Boards, seven Special NHS Boards, and one public health body,⁹⁷ and Scottish Water. These services rely entirely on reserved infrastructure networks: electricity grids, subsea fibre-optic cables, and digital telecommunication channels. Scottish Ministers lack primary legislative control over those networks. When the British Parliament introduced the Cyber Security and Resilience Bill to expand oversight across Managed Service Providers and data centres powering essential services,⁹⁸ it addressed digital vulnerabilities within a reserved regulatory domain that directly impacts devolved operational entities like NHS Scotland.⁹⁹ The Scottish Government acts as the competent regulatory authority for cyber resilience in devolved health and water sectors,¹⁰⁰ yet must align its policies with UK-wide frameworks established by the NCSC, leaving Scotland responsible for outcomes it cannot control.

Norway presents a unified structural counterpoint. Under the Norwegian Security Act, Sikkerhetsloven,¹⁰¹ Norway’s CNI governance integrates physical asset protection, subsea pipeline surveillance, electricity grid management, and cybersecurity under a single sovereign legal regime. Norway’s Ministry of Energy and Ministry of Justice and Public Security exercise direct authority over subsea gas pipelines operated by Gassco¹⁰² and the national power grid operated by Statnett.¹⁰³ As a European Economic Area member, Norway also implements the EU’s Network and Information Security Directive through a separate Digital Security Act, Digitalsikkerhetsloven, which operates alongside the Security Act.¹⁰⁴ Where both apply, the Security Act takes precedence as *lex specialis* on national security grounds. Norway’s legal architecture is layered, but it is coherent. One sovereign state holds ultimate authority over all dimensions of CNI governance.

China operationalises CNI governance through a hyper-centralised legislative and state apparatus anchored by the Cybersecurity Law, the Data Security Law, and the Regulation on Protecting the Security of Critical Information Infrastructure.¹⁰⁵ This regulatory triad grants the central state authority to audit, secure, and intervene in critical network operations. The State Grid Corporation serves as the principal operational entity for the national power grid.¹⁰⁶ This top-down structure reduces multi-jurisdictional consenting barriers and allows rapid deployment of physical security and digital monitoring across municipal and regional networks. Both

⁹⁷NHS Scotland (2022) “About NHS Scotland”.

⁹⁸UK Government (2026) “Cyber Security and Resilience (Network and Information Systems) Bill: Explanatory Notes”.

⁹⁹NHS National Services Scotland (2025) “Cyber Centre of Excellence hosts Open Day during CyberScotland week”.

¹⁰⁰Scottish Government (2026) “Legislative Consent Memorandum: Cyber Security and Resilience (Network and Information Systems) Bill”.

¹⁰¹Justis- og beredskapsdepartementet (2018) “Lov om nasjonal sikkerhet (sikkerhetsloven)”.

¹⁰²Gassco (2025) “Norne Gas Transport”.

¹⁰³Statnett (2025) “Introduction to our strategy”.

¹⁰⁴Nasjonal sikkerhetsmyndighet (2025) “Veileder fra NSM i digitalsikkerhetsloven og -forskriften”.

¹⁰⁵State Council of the People’s Republic of China (2021) “Regulation on Protecting the Security of Critical Information Infrastructure”.

¹⁰⁶State Grid Corporation of China (2025) “Stronger, Smarter, Greener: State Grid Powers Chinese Modernization”.

sovereign states achieve coherence through unified legal authority. Scotland's constitutional architecture distributes authority without a mechanism for integration, and the consequence is a governance gap that runs through every dimension of its CNI profile.

Conclusion

The finding is that CNI resilience depends on agency, and that coherence is one component of agency alongside fiscal capacity, technical capability, and operational competence. Norway and China achieve coherence through unified legal authority, and this contributes to their stronger CNI outcomes. But coherence is not sufficient. The Fosen ruling demonstrates that even a sovereign state with unified authority can produce outcomes that violate its own environmental and indigenous rights commitments. China's cyber workforce gap demonstrates that a state with full sovereign authority over education and industrial policy can still face capability limits that coherence cannot resolve. Scotland's CNI profile is therefore not simply a story of constitutional fragmentation. Instead it is logical to see it as a story of compounded agency gaps. Scotland lacks coherence because its authority is split, and it lacks the fiscal and technical capacity to compensate, because those instruments are held by the same authority that shapes the constitutional split. The gaps reinforce each other.

The dependency side of the gap is consistent across every PRESS dimension. Island communities cut off from severed cables, a trunk road unbuilt after fourteen years of assessment, an NHS that relies on networks Scottish Ministers cannot regulate, a cyber resilience framework determined by legislation passed elsewhere. The coherence side of the gap is constitutional. Scotland cannot close it unilaterally, and neither can Westminster, because the two authorities answer to different electorates with different priorities. That is what makes the vulnerability structural rather than contingent. It will remain so until the constitutional architecture changes.

Food Security

Food security is the biological foundation of Development-Security. It is not a measure of agricultural output. It is the structural capability of a society to feed its population with adequate, affordable, and ecologically resilient nutrition under conditions of external shock. Within the PRESS framework, food security sits at the intersection of natural resource management, land tenure, environmental sustainability, and sovereign regulatory authority. A secure development trajectory requires that primary food production does not deplete soil health, destroy biodiversity, or depend on hyper-globalised supply chains vulnerable to trade embargoes, climatic shocks, or foreign input price spikes.

Scotland presents a fundamental paradox. It possesses high-grade grazing land, world-class livestock production, vast marine protein capital, and net export capacity in major food categories. Its population nevertheless faces acute food insecurity and structural exposure to international supply chain shocks. This chapter argues that the paradox is not incidental, it is a structural condition produced by two distinct failures. The first is constitutional: the levers that determine food affordability, labour availability, and trade exposure remain reserved to Westminster, where they are exercised by an authority with no mandate over Scottish food outcomes. The second is domestic: even within devolved competence, land ownership concentration and subsidy design reproduce food insecurity that no constitutional settlement can resolve. The chapter's finding is therefore narrower than the previous two. Agency is necessary for food security, but not sufficient. Independence would remove one constraint without resolving the other. It is the insufficiency, not the necessity, that the independence debate has yet to confront.

Populations

The Populations dimension tests whether a food-producing nation can guarantee food access to its own population. An agricultural system that exports high-value produce while substantial segments of the domestic population endure food poverty represents a structural failure of distribution. Scotland demonstrates this failure precisely. The Scottish Health Survey 2024 found that 8% of adults worried about running out of food due to a lack of money in the previous twelve months, a decrease from 14% in 2023.¹⁰⁷ Relative poverty after housing costs stands at 17%, equivalent to 940,000 individuals.¹⁰⁸ Data from the Trussell network indicates that approximately 30% of people referred to food banks were in working households, up from 24% in 2022.¹⁰⁹ Food poverty is no longer confined to the unemployed but has become a condition the lower wages and rising cost of living.

The consequences of food illiquidity are unequally distributed across geography and demo-

¹⁰⁷Scottish Government (2025) "*The Scottish Health Survey 2024 - volume 1: main report*".

¹⁰⁸Scottish Government (2026) "*Poverty and income inequality in Scotland 2022-25*".

¹⁰⁹Trussell (2024) "*State of Hunger 2024*".

graphics. Urban post-industrial areas exhibit severe spatial deprivation. Remote rural and island communities face compromised food security due to long, fragile logistics corridors. High freight charges, ferry disruptions, and fuel price surcharges create a rural cost penalty that raises the cost of basic food baskets above Central Belt averages. The metaphorical geography of food poverty in Scotland is physical geography and access.

Norway faces the same external exposure as Scotland. It imports a significant share of its caloric intake and is subject to the same European price shocks. It does not translate those shocks into domestic food poverty, and the reason is instructive. Norway possesses two instruments Scotland lacks. The first is fiscal: during European inflation spikes, the Norwegian state leveraged petroleum revenues to cushion agricultural inputs and subsidise basic food supply chains, absorbing market volatility at the state level rather than passing it to households.¹¹⁰ The second is regulatory: Norway sets food pricing policy within a national framework, not a shared one. China manages population food security through direct state administration of staple grain availability and price stability. The Chinese central state enforces a minimum amount farmers get paid for their grain, which allows for the distribution of targeted input subsidies to rural smallholders, and for the state-directed retail logistics networks to guarantee that domestic price shocks do not translate into widespread hunger.¹¹¹ Therefore, unlike Scotland, both sovereign states treat food access as a state obligation that their institutions are equipped to deliver. The result is that 940,000 people in a food-exporting nation live in relative poverty.

Resources

The Resources dimension tests whether a state can convert agricultural abundance into food security. The test is not whether a country produces enough food, but whether it can guarantee the availability of that food under external shock. Scotland's resource profile appears abundant in aggregate and fragile in composition. Research by the Rowett Institute and SRUC confirms that Scotland is largely self-sufficient in cereals, potatoes, lamb, beef, dairy, and eggs, but has vulnerabilities in poultry and pork supply.¹¹² Self-sufficiency in low-value bulk commodities alongside import dependence in higher-value and horticultural categories reflects a sector shaped by comparative advantage rather than strategic provisioning. Less than 10% of Scottish agricultural land is prime arable land suitable for cash crops, with the vast majority classified as Rough Grazing and Less Favoured Areas suitable primarily for extensive livestock grazing.¹¹³ Scotland's caloric self-sufficiency is achieved through livestock production that itself depends on imported feed concentrates, synthetic fertilisers, and foreign machinery. Scotland's marine resource base partially offsets this terrestrial imbalance. Aquaculture produces almost 200,000 tonnes of Atlantic salmon annually.¹¹⁴ Marine protein wealth does not resolve the structural problem. It is export-oriented, priced by global markets, and does not substitute for the horti-

¹¹⁰Regjeringen (2026) "*Jordbruksavtalen styrker norsk matproduksjon i en urolig verden*".

¹¹¹中华人民共和国国家发展和改革委员会 (2026) "国家发展改革委等部门关于公布2026年稻谷最低收购价格的通知" -国家发展和改革委员会".

¹¹²Rathnayaka, S. D., C. Revoredo-Giha, and B. de Roos (2024) "'Assessing Scotland's self-sufficiency of major food commodities'".

¹¹³NatureScot (2023) "*Thematic document - arable*".

¹¹⁴The Scottish Government (2025) "*Scottish fish farm production survey 2024*".

cultural and poultry imports on which domestic food baskets depend. The result is a resource profile that appears abundant in aggregate and fragile in composition.

Norway presents a resource profile characterised by severe terrestrial constraints paired with immense marine wealth. Its cold climate and mountainous topography limit terrestrial food self-sufficiency to between 35% and 40% when adjusted for imported feed, or 40% and 45% unadjusted.¹¹⁵ Norway's marine sector produces over 1.6 million tonnes of farmed seafood annually, with production projected to grow to around 2.4 million tonnes by 2050.¹¹⁶ To address its terrestrial caloric deficit, Norway executed a strategic policy shift in June 2024 by re-establishing national emergency grain reserves. Citing the COVID-19 pandemic, war in Europe, and climate disruption, the Norwegian government signed contracts with private storage providers to stockpile 30,000 tonnes of food-grade wheat across 2024 and 2025. The state plans to expand these reserves to 82,500 tonnes by 2029, securing a continuous three-month strategic reserve of total caloric grain requirements for its 5.6 million citizens.¹¹⁷

China approaches agricultural resource security through absolute state control. The country enforces the statutory preservation of its 1.8 billion mu red line, equivalent to 120 million hectares of arable land, establishing an absolute minimum floor below which agricultural acreage cannot be converted to industrial or urban use. Through the state-owned China National Grain Reserve Corporation, or Sinograin, China maintains strategic stockpiles of wheat and rice estimated to cover over one year of full domestic consumption, insulating its 1.4 billion residents from global trade shocks.¹¹⁸ Both sovereign states treat food reserves as a strategic asset. Scotland cannot establish independent reserves because emergency reserve powers remain reserved to Westminster.

Ecosystems

The Ecosystems dimension tests whether agricultural development can be reconciled with ecological limits, and the answer depends on who holds the authority to change land use. Scotland's agricultural emissions profile reveals a structural contradiction at the heart of its climate strategy. Agriculture is a significant contributor to greenhouse gas emissions. Under the Scottish Climate Change Plan and NPF-4, the sector is simultaneously required to reduce emissions and expand its role as a biological carbon sink. This dual mandate is internally tensioned. It asks the same sector to shrink and grow at the same time.

Scotland contains over 2 million hectares of peatland, of which approximately 70% is degraded through drainage, overgrazing, and commercial forestry.¹¹⁹ Degraded peatland trans-

¹¹⁵Norwegian Institute of Bioeconomy Research (2026) “*Different calculation methods can lead to major variations in food self-sufficiency rates*”.

¹¹⁶Nordic Council of Ministers (2016) “*Marine aquaculture in the Nordics*”.

¹¹⁷Regjeringen (2024) “*Milepæl for norsk beredskap*”.

¹¹⁸中华人民共和国生态环境部 (2025) “第三场部长通道，说成绩话期待（现场·部长通道）_中华人民共和国生态环境部”.

¹¹⁹Scottish Government (2024) “*The Rural Support (Improvement) (Miscellaneous Amendment) (Scotland) Regulations 2024: Policy Note*”.

forms what should be a carbon sink into an active emissions source. The Scottish Government's commitment to restoring 250,000 hectares of degraded peatland by 2030¹²⁰ therefore requires more than just additional investment. It requires a fundamental restructuring of upland agricultural practice, including reduced stocking densities and altered grazing regimes. The political difficulty is that these changes impose concentrated costs on hill farming communities while delivering diffuse national benefits. The result is a climate strategy that depends on the voluntary cooperation of the very sector it seeks to transform.

Norway faces a different ecological constraint. Only 3% of its landmass is classified as cultivated agricultural land, which means soil conservation is regulated by necessity rather than choice.¹²¹ Norway's ecological pressures are concentrated in its marine sector. Intensive Atlantic salmon farming exerts pressure on coastal benthic ecosystems through organic waste, sea lice, and reliance on fishmeal inputs. The Norwegian response has been regulatory rather than redistributive. Environmental standards are imposed on operators, but the underlying structure of the industry remains unchanged. This distinguishes Norway from Scotland. Norwegian ecological reform requires operational compliance. Scottish ecological reform requires land use change, and land use change means confronting the ownership structures described in the Socio-Economy section.

China's agricultural ecology reflects decades of industrialisation paired with recent centralised remediation. Long-term overuse of synthetic fertilisers has caused soil acidification and groundwater pollution, creating an environment that now constrains yield growth.¹²² Under its "Ecological Protection Red Lines" framework, China has mandated reduced chemical application rates and constructed water-efficient irrigation infrastructure. The critical feature of this strategy is that it is enforced through the same top-down administrative apparatus that produced the original degradation. The state is both the cause of the ecological problem and the only actor capable of resolving it. China's ecological governance rests on a concentration of authority that Scotland does not possess. It also illustrates a limitation of that authority. The same centralised apparatus that produced soil degradation is the only actor capable of reversing it, which means the remedy depends on the state's willingness to correct itself. Agency concentrates the capacity for action. It does not determine the direction.

Socio-Economy

The Socio-Economy dimension tests whether constitutional authority alone can produce food security outcomes, or whether the domestic distribution of power within a policy domain also determines them. Scotland is the strongest test case for this question, because food policy, land reform, and subsidy design are all devolved. Scotland has the authority. Food insecurity has persisted. The dimension therefore reveals a different kind of agency failure than the previous two chapters examined. Not absence of authority, but insufficiency of it.

¹²⁰Scottish Government (2025) "*Record levels of peatland restored*".

¹²¹Statistisk sentralbyrå (2026) "*Land use and land cover*".

¹²²Zhu, X. et al. (2024) "'The contribution of natural and anthropogenic causes to soil acidification rates under different fertilization practices and site conditions in southern China'".

Land tenure and subsidy structures determine whether agricultural development strengthens rural communities or concentrates wealth. Scotland's land ownership pattern is among the most concentrated in Europe, with a small number of private landowners controlling a disproportionate share of rural land.¹²³ This concentration is not merely a legacy of feudal history as it still exists as a constraint of agricultural policy. It limits land access for young farmers, inflates rural property prices, and restricts the community-led development that might otherwise diversify rural economies. The Land Reform (Scotland) Act 2025 represents a deliberate attempt to disrupt this structure.¹²⁴ By introducing mandatory community engagement, pre-transfer community right to buy, and compulsory lotting controls for large holdings, the Act shifts the default from concentrated private control toward distributed community ownership. This is a structural intervention, and its significance lies in recognising that land tenure determines who benefits from agricultural policy. Crucially, land reform is a devolved competency. The Scottish Parliament possesses the authority to legislate on this. The persistence of concentrated land ownership is therefore a domestic political outcome, not a constitutional constraint.

Scotland's post-Brexit subsidy transition compounds this challenge. Under the Agriculture and Rural Communities (Scotland) Act 2024, Holyrood is phasing out unconditioned direct payments in favour of a framework linking public support to eco-conditions and biodiversity net gain.¹²⁵ The shift is progressive in environmental terms and regressive in distributional terms. It rewards landowners who can afford the transition costs and penalises those who cannot. This too is a devolved policy choice. Independence would not alter the trade-offs it embodies.

Norway's socio-economic model operates on different assumptions. Its agricultural framework is explicitly protectionist, with among the highest levels of OECD support and a Producer Support Estimate averaging 47% of gross farm receipts in 2022 to 2024.¹²⁶ The Jordbruksavtalen system is not designed to maximise efficiency. It is designed to preserve rural settlement patterns and family farm structures, which results in high consumer prices and inefficient production, but also stable rural communities and reduced depopulation. Norway has chosen this outcome through sovereign policy.

China's model rests on collective rural land ownership and the Household Responsibility System, under which rural households hold long-term use rights while ownership remains with the village collective. This arrangement prevents the emergence of a landed class and preserves state control over land allocation. The state then funds rural infrastructure, guarantees minimum support prices for grains, and provides targeted credit through the Agricultural Development Bank of China, sustaining hundreds of millions of smallholder livelihoods. The system is inefficient in market terms. It is politically stabilising in development-security terms. The comparison with Scotland is instructive. China's rural land system is a sovereign policy choice. Scotland's con-

¹²³Scottish Government (2016) "*The impact of diversity of ownership scale on social, economic and environmental outcomes.*"

¹²⁴Scottish Land Commission (2025) "*Land Reform (Scotland) Act 2025*".

¹²⁵Scottish Government (2024) "*Agriculture and Rural Communities (Scotland) Act 2024*".

¹²⁶OECD (2024) "*Agricultural Policy Monitoring and Evaluation 2024: Innovation for Sustainable Productivity Growth*".

centrated land ownership is also a policy choice. Both are made within different constitutional frameworks, but neither is dictated by the constitution itself.

Structures

Scotland's food security governance operates under an asymmetric constitutional arrangement. Agriculture, food policy, spatial planning, and land reform are devolved competencies fully within the Scottish Parliament's jurisdiction. Key devolved frameworks include the Good Food Nation (Scotland) Act 2022, the Agriculture and Rural Communities (Scotland) Act 2024, and the Land Reform (Scotland) Act 2025. The Scottish Parliament can legislate on food policy. It can reform land tenure. It can set agricultural subsidy conditions. These are substantial powers.

The problem is that key macro-economic levers fundamental to long-term food security remain reserved to Westminster:

- **International Trade:**¹²⁷ Post-Brexit Free Trade Agreements with Australia and New Zealand have introduced tariff-free import quotas for foreign beef and lamb. Scottish agricultural bodies argue these agreements undercut domestic producers operating under stricter environmental and welfare regulations.
- **Immigration and Seasonal Labour:** Regulation of foreign labour access is reserved to the UK Home Office. Following Brexit, Scottish soft fruit and seafood processing sectors have faced seasonal labour shortages.
- **UK Internal Market Act 2020:**¹²⁸ Mutual recognition principles prevent Holyrood from banning the sale of food goods produced in England that do not meet Scottish standards. The Act disappplies devolved regulatory requirements in relation to goods and services from other parts of the UK, limiting the effectiveness of Scottish legislation even where devolved competence is formally unchanged.
- **Emergency Reserve Powers:** Under UKIMA, as well as the Scotland Act which retains emergency-level powers to the House of Commons, Scotland lacks independent authority to establish national strategic grain reserves without UK-wide coordination.

The list is not a list of grievances, but instead a list of instruments that determine food security outcomes, none of which Scotland controls. The devolved institutions are responsible for outcomes they lack the instruments to deliver. Where the Energy chapter showed partial agency over resources, and the CNI chapter showed fragmented agency over infrastructure, food security shows a third pattern: Misaligned agency. Scotland holds authority over some of the means of food security and none of the means of food affordability.

Norway operates a streamlined, sovereign governance structure anchored in the Ministry of

¹²⁷Centre for Inclusive Trade Policy (2025) “Copy, paste, repeat: The trend and risks of boilerplating in UK Free Trade Agreements”.

¹²⁸Brown Swan, C. (2026) “Explainer: The UK Internal Market Act 2020 and implications for the Scottish Government”.

Agriculture and Food. While Norway is an EEA member, it negotiated complete exemptions from the EU's Common Agricultural Policy and Common Fisheries Policy. This permits Norway to enforce independent import tariffs, maintain national biosecurity mandates, and execute national strategic grain stockpiling decisions.¹²⁹ Norway's sovereignty is not absolute as it is bounded by EEA membership, but it is sufficient to protect national food security as a strategic objective.

China operates under a centralised legislative apparatus. On 1st June 2024, China enacted its Food Security Law.¹³⁰ This comprehensive statute codifies absolute state authority over the entire agricultural value chain. It establishes statutory penalties for local officials who fail to protect designated prime farmland under the 1.8 billion mu red line. It requires provincial governments to maintain specific grain self-sufficiency ratios. It mandates state oversight and investment in domestic seed genetics. It grants the central state power to execute price controls, ration key inputs, and requisition storage assets during national food supply disruptions. China's Food Security Law is the most explicit codification of food security as a sovereign obligation in the contemporary world. Both Norway and China treat food security as a state responsibility. Scotland treats it as a policy domain distributed across two governments, neither of which possesses the full set of instruments required to guarantee domestic nutrition.

Conclusion

The comparative evidence across all five PRESS dimensions produces a finding that complicates the argument of the previous two chapters. Food policy is devolved. Land reform is devolved. Subsidy design is devolved. The Scottish Parliament possesses the authority to re-structure land ownership, redirect agricultural subsidies, and reshape rural economies, and it has exercised that authority. The Land Reform (Scotland) Act 2025 and the Agriculture and Rural Communities (Scotland) Act 2024 are both significant interventions. Food insecurity has persisted despite them. This suggests that agency is necessary for food security, but not sufficient. Independence would remove the constitutional constraint identified in this chapter. It would not resolve the domestic structural constraints identified in the Socio-Economy dimension. Land ownership concentration, subsidy design, and rural poverty are policy outcomes, and policy outcomes are matters of political will which cannot be settled by a constitutional referendum.

Norway and China present contrasting models of sovereign food governance. Norway combines extreme agricultural protectionism with marine resource wealth and a re-established strategic grain reserve. China enforces caloric self-sufficiency through statutory grain redlines and state-owned reserves. Both models prioritise national food security as a sovereign obligation. Both operate within constitutional frameworks that permit integrated governance across trade, labour, land, and emergency reserves. Scotland does not. The comparative evidence suggests that food security resilience depends on the constitutional architecture that shapes who controls

¹²⁹Confederation of British Industry (2022) "*Business view of the EU from Norway*".

¹³⁰国家粮食和物资储备局 (2023) "中华人民共和国粮食安全保障法".

trade, labour, and emergency reserves. Where that architecture is fragmented, domestic abundance does not guarantee domestic nutrition, but where that architecture is unified, domestic policy choices still determine who eats and who does not. Agency creates the conditions for food security, it cannot guarantee it. That is the harder lesson of this chapter, and it is the lesson that the independence debate has yet to confront.

Part III

Analysis

The Independence Variable

The three empirical chapters established that Scotland's Development-Security profile is shaped by the distribution of agency, and that the pattern of failure is not uniform. Energy shows agency absent where it most matters, reserved competencies prevent Scotland from capturing the benefits of its resources. CNI shows agency fragmented across reserved and devolved domains, compounding with fiscal and technical gaps. Food Security shows agency present and insufficient: policy is devolved, authority is exercised, and food insecurity persists. This chapter isolates the independence variable by asking how independence would reconfigure each pattern. Its finding is that independence resolves the constitutional failures and leaves the political ones intact. The necessity may be constitutional but the insufficiency is purely political. The analysis draws on the Scottish Government's *Building a New Scotland* series as evidence of what independence could enable.¹³¹

How Independence Reconfigures Agency

In energy, independence would resolve the absence of agency identified in Chapter 4. It would transfer control over wholesale market design, offshore licensing, and transmission tariff regulation to a sovereign Scottish state, giving Scotland the instruments to decouple domestic consumer tariffs from international gas prices, attach binding local content requirements to licensing, and negotiate interconnection terms as a counterparty rather than a subordinate jurisdiction. Independence would give Scotland the legal instruments to design its own energy framework. It would not supply the institutional infrastructure to operate them, which would have to be built, staffed, and integrated with the systems Scotland currently shares.

In terms of CNI, Scottish independence could allow for a greater degree of autonomy required to integrate physical and digital systems, as well as increase CNI resilience across the nation. This can be seen in the "National Islands Plan", where the government has already aligned itself to increase transport and productivity.¹³² Norway's unified Security Act and China's centralised regulatory triad provide contrasting models: coherence is not a property of sovereignty but of the constitutional design that sovereignty permits. But coherent governance is not immune to failure. The Fosen ruling showed that Norway's unified legal authority produced an outcome that violated its own commitments. Scotland's fragmented system has the opposite problem: it can produce failures that neither authority has the mandate or the instruments to correct. Independence would replace one set of problems with another.

In food security, the 8th paper released by the Scottish government centred on the marine sector and the possible commitments the Scottish government may enact once it is exempt

¹³¹Scottish Government (2024) "*Building a New Scotland*".

¹³²Scottish Government (2026) "*National Islands Plan*".

from the UK Internal Market. These provisions include rejoining the European market as 64% of Scottish salmon (the UK's "biggest food export") is already destined for European markets.¹³³ The domestic structural constraints (land ownership concentration, subsidy design, rural poverty) are matters of political will, and political will is not a constitutional variable. Independence would expand what Scotland can do but could not change what Scotland has done.

What Independence Would Not Change

Three constraints would persist, and they are of different kinds. The material constraint is geography. Subsea cables and strategic transport corridors remain vulnerable to disruption, and independence changes who is responsible for managing them, not the vulnerabilities themselves. The structural constraint is supply chain dependence. Scotland's renewable energy sector, food sector, and infrastructure depend on international manufacturers and global markets that a small state cannot substitute at scale. The geopolitical constraint is the international order. Scotland would join the community of states as a small power with limited reach, and formal sovereignty would not translate into the geopolitical weight Norway and China exercise. None of these are arguments against independence. They are arguments against treating independence as sufficient. Norway's institutional infrastructure was built over decades. China's cyber workforce gap persists despite full state control.

European Accession

Independence is not the only constitutional variable in play. Scotland's route would likely involve EU accession, and accession would interact with each domain differently.

In energy, EU membership would expand the agency that independence creates. Scotland's renewable generation exceeds domestic demand, and the constraint on realising its value is market access. Independence would give Scotland control over licensing and market design, but it would not create demand for the electricity Scotland generates. EU membership would give Scotland access to: the internal energy market (which provides access to a continental market of 450 million consumers), interconnection rights, and EU funding mechanisms from which the UK has been excluded since Brexit. Independence supplies the authority; accession supplies the market and the capital. Bridging the 28.7% fuel poverty rate would require both at the same time.

In CNI, EU membership would not expand Scottish agency so much as change the terms of the Scotland-UK relationship. Post-independence, shared infrastructure would become cross-border infrastructure between an EU member state and a third country. The EU regulates third-country access with stricter security and resilience standards than the UK currently applies. Scotland would be the beneficiary of those standards, and the UK would be subject to them. The governance gap identified in Chapter 5 would not disappear. It would be reconfigured, and

¹³³Scottish Government (2023) "*Our marine sector in an independent Scotland*".

Scotland would be the stronger party in the cross-border relationship..¹³⁴ This however ignores the damage EU accession may play on cross-border communities, which is riddled with so many variables that cannot yet be remotely simulated. The Scottish Government ultimately dictates the quality of life for these communities, which rely heavily on accessing services and CNI from both sides of the arbitrary divide.

In food security, EU membership would reconfigure agency completely. Scotland would gain single market access for food and drink exports and integration into European research networks. It would also operate under the Common Agricultural Policy and Common Fisheries Policy, and its third-country trade policy would be set by Brussels. Norway holds complete exemptions from both frameworks and uses them to maintain agricultural protection that would not be permissible under EU rules. Scotland would not have that option. Post-accession Scotland would have more agency over land reform, subsidy design, and rural development, and less over agricultural protection, fisheries, and trade. Independence from Westminster expands the toolkit while EU accession would partially constrain it again.¹³⁵ **The Independence Variable**

The comparative evidence produces a specific finding. Independence resolves the constitutional constraints and leaves the political ones intact. EU accession then introduces a second set of constraints, expanding Scottish agency in energy, strengthening it in CNI, and partially constraining it in food. The sharpest version of the finding is that independence is necessary for the resolution of constitutional agency failures, but not sufficient, and EU accession adds a second layer of constraint that the independence debate has not yet acknowledged.

The comparative cases confirm the pattern. Norway converts formal authority into material outcomes because the institutional infrastructure exists, built over decades of sovereign statehood. China's state capacity permits centralised action across the value chain, and it still faces capability gaps that authority alone cannot close. Scotland would enter independence with neither Norway's institutional infrastructure nor China's scale. It would have to build the infrastructure that converts agency into outcomes. Independence without that infrastructure would not produce the Norwegian model. It would produce a state with formal sovereign authority and insufficient institutional capacity to exercise it.

¹³⁴European Union (2022) "*Directive 2022/2557*".

¹³⁵European Union (2024) "*Acquis*".

Conclusion

Scotland's Development-Security profile is uniquely paradoxical, because it is not a case of developmental failure. Both the capacity and the capability to grow in a sustainable manner are present. The agency required to act on them is not. Across the three empirical domains, agency fails in three distinct ways, and each produces a different profile. In energy, agency is absent where it matters most. Scotland generates more renewable electricity than it consumes and cannot protect its households from the price consequences of a market it does not design. In critical national infrastructure, agency is fragmented. Authority is split across reserved and devolved matters, and the gaps between them compound with fiscal and technical shortfalls rather than sitting alongside them. In food security, agency is present but insufficient. The Scottish Parliament holds the relevant competences, and yet food insecurity persists.

The three modes are not variations on a single failure. Each identifies a different capacity within agency itself. In energy, the missing capacity is constitutional: the authority to design markets, set tariffs, and license resources sits elsewhere, so the resources Scotland holds cannot be converted. In critical national infrastructure, the missing capacity is integrative: authority exists on both sides of the constitutional divide, but nothing connects them, and the gap compounds with fiscal and technical shortfalls. In food security, the missing capacity is distributive: the competences are held, exercised, and insufficient, because the instruments that determine affordability and access sit outside them. Agency is therefore not a single variable but a family of capacities, and the shape of each failure identifies which member of the family is absent. This is the answer to the research question. The distribution of agency does not simply determine whether Scotland succeeds; it determines how Scotland fails, and it does so in a pattern that is consistent across three unrelated policy domains.

The question posed was simple: how does the distribution of agency shape Scotland's Development-Security profile in comparison to Norway and China? The answer is that the distribution determines which capacities are available, which are absent, and therefore which failure modes are possible. Norway and China do not outperform Scotland because they possess resources Scotland lacks, or because they are more committed to sustainable development. They outperform Scotland because they hold the full set of capacities required to convert what they have into what they need. The international system was built on the claim that sovereignty is the price of statehood, and that those without it lack the ability to secure themselves. Sovereignty was always one configuration of agency rather than its definition, and Scotland makes that visible. Scotland is not a failed case of statehood; it is a functioning nation that grows, exports, and legislates. But it cannot secure the conditions of its own development, because the authority to convert its capacities into outcomes is not its own. The premise was never that sovereignty guarantees security, but that its absence guarantees insecurity. That half holds. The other, less so. The reason? **Agency.**

Part IV

Reference List

Reference List

- [1] Arvesen, A. et al. (2025) ““On the potential role of flexible Norwegian hydropower in managing challenging renewable energy variability events in Europe””. In: *IOP Conference Series: Earth and Environmental Science* 1442.1 (), p. 012003. DOI: [10.1088/1755-1315/1442/1/012003](https://doi.org/10.1088/1755-1315/1442/1/012003) (cited on page 17).
- [2] Barnes, Bräuer, and Tahir (2024) “*GB wholesale electricity market reform: Impacts and opportunities for scotland*”. Tech. rep. Energy Reform Agency. URL: <http://dx.doi.org/10.7488/era/4431> (cited on page 16).
- [3] BBC News (2020) ““Significant’ landslide forces closure of rest and be thankful””. In: *BBC News* (). URL: <https://www.bbc.co.uk/news/uk-scotland-glasgow-west-54137275> (cited on page 21).
- [4] British Government (2009) “*Climate Change (Scotland) Act 2009, Section 53: Programmes for adaptation to climate change*”. URL: <https://www.legislation.gov.uk/asp/2009/12/section/53/enacted> (cited on page 23).
- [5] British Government (1989) “*Electricity Act 1989*”. Sections 36 and 37 govern Scottish Ministers’ consenting powers for onshore electricity infrastructure. URL: <https://www.legislation.gov.uk/ukpga/1989/29/contents> (cited on page 24).
- [6] British Government (1998) “*Scotland Act 1998*”. URL: <https://www.legislation.gov.uk/ukpga/1998/46/contents> (cited on page 19).
- [7] Brown Swan, C. (2026) “*Explainer: The UK Internal Market Act 2020 and implications for the Scottish Government*”. Tech. rep. SB 26-43. Scottish Government Information Centre (SPICe). URL: <https://www.parliament.scot/chamber-and-committees/research-prepared-for-parliament/research-briefings/2026/7/8/sb-2643/pdf> (cited on page 34).
- [8] Buzan, B., O. Wæver, and J. De Wilde (1998) “*Security: A new framework for Analysis*”. Colorado: Lynne Rienner Publishers (cited on pages 5, 8).
- [9] Centre for Inclusive Trade Policy (2025) “*Copy, paste, repeat: The trend and risks of boilerplating in UK Free Trade Agreements*”. University of Sussex. URL: <https://citp.ac.uk/publications/copy-paste-repeat-the-trend-and-risks-of-boilerplating-in-uk-free-trade-agreements> (cited on page 34).
- [10] CGTN (2022) “*China’s demand for cybersecurity talents to reach 3.27 million By 2027*”. URL: <https://news.cgtn.com/news/2022-09-08/China-s-demand-for-cybersecurity->

- talents-to-reach-3-27-million-by-2027-1dakC99UZws/index.html (cited on page 26).
- [11] CGTN (2026) ““China’s west-to-East power project delivers over 1 trillion kWh to Jiangsu””. In: *CGTNOfficial* (). URL: <https://news.cgtn.com/news/2026-03-20/China-s-west-to-east-power-project-tops-1-trillion-kWh-to-Jiangsu-1LFIw1iy1t6/p.html> (cited on page 23).
- [12] Chinese Academy of Sciences (2025) “*New Study Reveals China’s Ecological Red Lines Offer Blueprint for Biodiversity Protection*”. URL: https://english.cas.cn/newsroom/research_news/life/202507/t20250722_1048144.shtml (cited on page 24).
- [13] 中华人民共和国国家发展和改革委员会 (2025) “关于促进电网高质量发展的指导意见(发改能源〔2025〕1710号) -国家发展和改革委员会”. Ndrc.gov.cn. URL: https://www.ndrc.gov.cn/xxgk/zcfb/tz/202512/t20251231_1402949.html (visited on 09/19/2026) (cited on page 19).
- [14] 中华人民共和国国家发展和改革委员会 (2026) “国家发展改革委等部门关于公布2026年稻谷最低收购价格的通知 -国家发展和改革委员会”. Ndrc.gov.cn. URL: https://www.ndrc.gov.cn/fzggw/jgsj/jgs/sjdt/202604/t20260402_1404481.html (visited on 09/19/2026) (cited on page 30).
- [15] 中华人民共和国生态环境部 (2025) “第三场部长通道，说成绩话期待（现场·部长通道） -中华人民共和国生态环境部”. Mee.gov.cn. URL: https://www.mee.gov.cn/zcwj/zclcfh/202503/t20250312_1103869.shtml (cited on page 31).
- [16] 国家粮食和物资储备局 (2023) “中华人民共和国粮食安全保障法”. Food Security Law of the People’s Republic of China, adopted 29 December 2023, effective 1 June 2024. URL: https://www.lswz.gov.cn/html/zcfb/2023-12/30/content_278749.shtml (cited on page 35).
- [17] 新浪财经 (2026) ““国网甘肃物资事业部多措并举护航重点项目建设””. In: 新浪财经 (). URL: <https://finance.sina.com.cn/jjxw/2026-06-25/doc-inieqnaz7820362.shtml> (cited on page 26).
- [18] Climate Change Committee (2025) “*Progress in reducing emissions in Scotland 2025 report to Parliament*”. Tech. rep. URL: <https://www.theccc.org.uk/wp-content/uploads/2026/02/Progress-reducing-emissions-in-Scotland-2025-report-to-Parliament-34378gfhdk928365sjfge.pdf> (visited on 09/19/2026) (cited on page 17).
- [19] Confederation of British Industry (2022) “*Business view of the EU from Norway*”. Tech. rep. URL: <https://www.cbi.org.uk/media/1326/view-of-the-eu-from-norway.pdf> (visited on 09/19/2026) (cited on page 35).

- [20] Cyberspace Administration of China (PRC) (2016) “中华人民共和国电信条例”. Tech. rep. 中央网络安全和信息化委员会办公室. URL: https://www.cac.gov.cn/2000-09/30/c_133142151.htm (cited on page 22).
- [21] Dalby, S. (2013) “*Security and environmental change*”. John Wiley & Sons (cited on page 5).
- [22] Department for Energy Security and Net Zero (2026) “*National Policy Statement for Electricity Networks Infrastructure (EN-5)*”. Applies to England and Wales only; use for comparative landscape/visual impact context. URL: <https://www.gov.uk/government/publications/national-policy-statement-for-electricity-networks-infrastructure-en-5-2025> (cited on page 24).
- [23] Drax Group (2026) “*Pumped Storage Hydro*”. URL: <https://www.drax.com/what-we-do/pumped-storage-hydro/> (cited on page 17).
- [24] Duffield, M. (2014) “*Global Governance and the New Wars: The Merging of Development and Security*”. 2nd. London: Bloomsbury Publishing (cited on pages 3, 5–7).
- [25] Dunshea, J. (2024) “*Decision eastern green link 1 - project assessment*”. Tech. rep. OFGEM. URL: <https://www.ofgem.gov.uk/sites/default/files/2024-11/EGL1%20Project%20Assessment%20Decision%20Final.pdf> (cited on page 22).
- [26] Dyet, G. et al. (2022) “*A83 rest and be thankful story map*”. URL: <https://storymaps.arcgis.com/collections/927aafb643374492b70b71408b8e9508> (cited on page 21).
- [27] Eastern Green Link 1 (2023) “*About EGL1*”. URL: <https://www.easterngreenlink1.co.uk/about-the-project> (cited on page 22).
- [28] Eastern Green Link 2 (2024) “*Go-ahead for UK’s biggest subsea connection project*”. URL: <https://www.easterngreenlink2.co.uk/article/go-ahead-uks-biggest-subsea-connection-project> (cited on page 22).
- [29] Edinburgh Napier (2024) “*Edinburgh Napier recognised as an academic centre of excellence in cyber security education*”. URL: <https://www.napier.ac.uk/about-us/news/edinburgh-napier-recognised-as-an-academic-centre-of-excellence-in-cyber-security-education> (cited on page 26).
- [30] Energy Action Scotland (2024) “*Fuel poverty*”. EAS. URL: <https://eas.org.uk/fuel-poverty/> (visited on 09/19/2026) (cited on page 16).
- [31] European Union (2024) “*Acquis*”. eur-lex.europa.eu. URL: <https://eur-lex.europa.eu/EN/legal-content/glossary/acquis.html> (visited on 09/19/2026) (cited on page 40).
- [32] European Union (2022) “*Directive 2022/2557*”. Europa.eu. URL: <https://eur-lex.europa.eu/eli/dir/2022/2557/oj/eng> (visited on 09/19/2026) (cited on page 40).

- [33] Gassco (2025) “*Norne Gas Transport*”. URL: <https://gassco.eu/en/rorledningsnettverk/norne-gas-transport/> (cited on page 27).
- [34] Giddens, A. (1984) “*The constitution of society: Introduction of the theory of structuration*”. Berkeley: University Of California Press (cited on page 7).
- [35] Global Energy Monitor (2024) “*China continues to lead the world in wind and solar, with twice as much capacity under construction as the rest of the world combined*”. Global Energy Monitor. URL: <https://globalenergymonitor.org/research/china-continues-lead-world-wind-and-solar-twice-much-capacity-under-construction-rest> (visited on 09/19/2026) (cited on page 18).
- [36] Government, N. (2025) “*Revidert nasjonalbudsjett: Norgespris pada straum og fjernvarme*”. Tech. rep. Regjeringa.no. URL: <https://www.regjeringen.no/no/aktuelt/norgespris-pa-strom-og-fjernvarme/id3100201/> (cited on page 16).
- [37] Hakimian, R. (2026) ““£3Bn of contracts signed on Fourth England to Scotland electricity ‘superhighway’””. In: *New Civil Engineer* (). URL: <https://www.newcivilengineer.com/latest/3bn-of-contracts-signed-on-fourth-england-to-scotland-electricity-superhighway-30-03-2026/> (cited on page 22).
- [38] Homer-Dixon, T. F. (1999) “*Environment, scarcity, and violence*”. Princeton, N.J; Oxford, England: Princeton University Press (cited on page 5).
- [39] Imperial Consultants (2025) “*Dogger Bank Final Report: HVDC Supply Chain and Procurement*”. Tech. rep. Imperial Consultants. URL: <https://www.imperial-consultants.co.uk/wp-content/uploads/2025/06/Dogger-Bank-Final-Report-16-April-2025.pdf> (cited on page 25).
- [40] International Energy Agency (2025) “*Building the Future Transmission Grid: Strategies to Navigate Supply Chain Challenges*”. Tech. rep. IEA. URL: <https://www.iea.org/reports/building-the-future-transmission-grid> (cited on page 25).
- [41] International Energy Agency (2026) “*Norway - countries & regions*”. URL: <https://www.iea.org/countries/norway/energy-mix> (cited on page 16).
- [42] Jenkins, K. et al. (2016) ““Energy justice: A conceptual review””. In: *Energy Research & Social Science* 11 (), pp. 174–182. DOI: [10.1016/j.erss.2015.10.004](https://doi.org/10.1016/j.erss.2015.10.004) (cited on pages 5, 8).
- [43] Jolly, R. and D. Ray (2006) “*The human security framework and national human development reports: A review of experiences and current debates*”. Tech. rep. United Nations. URL: <https://hdr.undp.org/system/files/documents/human-security.pdf> (cited on page 3).

- [44] Justis- og beredskapsdepartementet (2018) “*Lov om nasjonal sikkerhet (sikkerhetsloven)*”. National Security Act. URL: <https://lovdata.no/dokument/NL/lov/2018-06-01-24> (cited on page 27).
- [45] Klaver, N. R. et al. (2024) ““Mass movement hazard & global climate change: Physical-deterministic modelling study of the Rest and be Thankful pass, Scotland UK””. In: *Geomorphology* 446 (), p. 108979. DOI: [10.1016/j.geomorph.2023.108979](https://doi.org/10.1016/j.geomorph.2023.108979). URL: <https://nora.nerc.ac.uk/id/eprint/536767/> (cited on page 23).
- [46] Klima- og miljødepartementet and Kommunal- og distriktsdepartementet (2017) “*Forskrift om konsekvensutredninger*”. FOR-2017-06-21-854, sist endret FOR-2026-03-06-353. URL: <https://lovdata.no/dokument/SF/forskrift/2017-06-21-854> (cited on page 24).
- [47] Lippuner, R. and B. Werlen (2009) ““Structuration theory””. In: *Elsevier eBooks*, pp. 39–49. DOI: [10.1016/b978-008044910-4.00749-5](https://doi.org/10.1016/b978-008044910-4.00749-5) (cited on page 5).
- [48] Liu, S., W. Ye, et al. (2026) ““Evaluating and optimizing residential electricity price tiers considering income redistribution equity under cross-subsidies mechanisms””. In: *Energies* 19.12 (), p. 2774. DOI: [10.3390/en19122774](https://doi.org/10.3390/en19122774) (cited on page 16).
- [49] Met Office (2019) “*UKCP18 Factsheet: Scotland*”. Tech. rep. Met Office Hadley Centre. URL: <https://www.metoffice.gov.uk/binaries/content/assets/metofficegovuk/pdf/research/ukcp/ukcp18-factsheet-scotland.pdf> (cited on page 23).
- [50] Ministry of Ecology and Environment (PRC) (2019) “国务院新闻办公室发布《碳达峰碳中和的中国行动》白皮书 (附全文)”. Tech. rep. 中华人民共和国生态环境部. URL: https://www.mee.gov.cn/ywgz/xdqhbh/qhbhlf/202511/t20251108_1132414.shtml (cited on page 18).
- [51] Nasjonal sikkerhetsmyndighet (2025) “*Veileder fra NSM i digitalsikkerhetsloven og -forskriften*”. On the relationship between the Security Act and the Digital Security Act. URL: <https://nsm.no/getfile.php/1314788-1759304646/NSM/Filer/Dokumenter/Veiledere/Veileder%20fra%20NSM%20i%20digitalsikkerhetsloven%20og%20-forskriften.pdf> (cited on page 27).
- [52] National Cyber Security Centre (2025) “*The NCSC at a glance*”. URL: <https://www.ncsc.gov.uk/collection/ncsc-annual-review-2025/ncsc-glance> (cited on page 26).
- [53] National Development and Reform Commission (2026) “新型能源体系建设‘十五五’规划”. Tech. rep. State Council of the People’s Republic of China. URL: <https://www.ndrc.gov.cn/xxgk/zcfb/tz/202606/P020260625612978907651.pdf> (cited on page 17).
- [54] National Protective Security Authority (2026) “*Critical National Infrastructure*”. URL: <https://www.npsa.gov.uk/about-npsa/critical-national-infrastructure> (cited on page 26).

- [55] NatureScot (2023) “*Thematic document - arable*”. NatureScot. URL: <https://www.nature.scot/doc/thematic-document-arable> (cited on page 30).
- [56] NESO (2022) “*Scottish boundaries — national energy system operator*”. URL: <https://www.neso.energy/publications/electricity-ten-year-statement-etys/electricity-transmission-network-requirements/scottish-boundaries> (cited on page 22).
- [57] NHS National Services Scotland (2025) “*Cyber Centre of Excellence hosts Open Day during CyberScotland week*”. URL: <https://www.nss.nhs.scot/news/cyber-centre-of-excellence-hosts-open-day-during-cyberscotland-week/> (cited on page 27).
- [58] NHS Scotland (2022) “*About NHS Scotland*”. URL: <https://www.scot.nhs.uk/about-nhs-scotland/> (cited on page 27).
- [59] Nordic Council of Ministers (2016) “*Marine aquaculture in the Nordics*”. Tech. rep. Norden.org. URL: <https://pub.norden.org/nordicinnovation2025-05/5-marine-aquaculture-in-the-nordics.html> (cited on page 31).
- [60] Norges Bank Investment Management (2026) “*Half-year report 2026: Government Pension Fund Global*”. Tech. rep. Norges Bank Investment Management. URL: <https://prod.generasjonsfondet.no/en/news-and-insights/reports/2026/half-year-report-2026/web-report-half-year-report-2026/> (cited on page 26).
- [61] Norges Bank Investment Management (2019) “*Oljefondet*”. URL: <https://www.nbim.no/no/> (cited on page 18).
- [62] Norges Domstoler (2021) “*Supreme Court of Norway, Statnett SF et al. v. Sør-Fosen sjte*”. URL: <https://www.domstol.no/globalassets/upload/hret/avgjorelser/2021/oktober-2021/hr-2021-1975-s.pdf> (cited on page 18).
- [63] Norges institusjon for menneskerettigheter (2024) “*Evalueringen av Fosen-saken*”. URL: <http://www.nhri.no/2024/evalueringen-av-fosen-saken/> (cited on page 24).
- [64] Norges institusjon for menneskerettigheter (2025) “*Fosen-aksjonistene vant frem i Høyesterett*”. URL: <https://www.nhri.no/2025/fosen-aksjonistene-vant-frem-i-hoyesterett/> (cited on page 24).
- [65] Norwegian Institute of Bioeconomy Research (2026) “*Different calculation methods can lead to major variations in food self-sufficiency rates*”. URL: <https://www.nibio.no/en/news/2026/different-calculation-methods-can-lead-to-major-variations-in-food-self-sufficiency-rates> (cited on page 31).
- [66] Norwegian Petroleum Safety Authority (2026) “*Integrating safety and security*”. URL: <https://www.havtil.no/en/explore-technical-subjects2/technical-competence/features/2024/integrating-safety-and-security/> (cited on page 23).

- [67] OECD (2024) “*Agricultural Policy Monitoring and Evaluation 2024: Innovation for Sustainable Productivity Growth*”. Tech. rep. OECD Publishing. DOI: [10.1787/74da57ed-en](https://doi.org/10.1787/74da57ed-en). URL: <https://doi.org/10.1787/74da57ed-en> (cited on page 33).
- [68] OFGEM (2024) “*Proposed anglo-scottish electricity superhighway is first to clear final fast-track funding hurdle*”. URL: <https://www.ofgem.gov.uk/press-release/proposed-anglo-scottish-electricity-superhighway-first-clear-final-fast-track-funding-hurdle> (cited on page 22).
- [69] OHCHR (1985) ““International Covenant on civil and political rights””. In: *Human Rights Quarterly* 7.1 (), p. 132. DOI: [10.2307/762040](https://doi.org/10.2307/762040) (cited on page 18).
- [70] Padley, M., A. Bryan, et al. (2026) “*The cost of remoteness: Reflecting higher living costs in remote rural Scotland when measuring fuel poverty 2024 update*”. Tech. rep. Scottish Government. URL: <https://www.gov.scot/binaries/content/documents/govscot/publications/research-and-analysis/2026/06/cost-remoteness-reflecting-higher-living-costs-remote-rural-scotland-measuring-fuel-poverty-2024-update/documents/cost-remoteness-reflecting-higher-living-costs-remote-rural-scotland-measuring-fuel-poverty-2024-update/cost-remoteness-reflecting-higher-living-costs-remote-rural-scotland-measuring-fuel-poverty-2024-update/govscot%3Adocument/cost-remoteness-reflecting-higher-living-costs-remote-rural-scotland-measuring-fuel-poverty-2024-update.pdf> (cited on page 16).
- [71] People’s Daily (2025) ““China puts 45 ultra-high-voltage projects into operation””. In: *People.cn* (). URL: <https://en.people.cn/n3/2025/1030/c90000-20383967.html> (cited on page 23).
- [72] Pizzuto-Pomaco, J. (2025) ““Internet cut off on Tiree after Storm Amy damages cable””. In: *The Herald* (). URL: <https://www.heraldscotland.com/news/25529503.internet-cut-off-tiree-storm-amy-damages-cable/?ref=cprfa#1> (cited on page 20).
- [73] PwC (2026) “*Norway - Petroleum tax regime*”. URL: <https://taxsummaries.pwc.com/norway/corporate/taxes-on-corporate-income> (cited on page 18).
- [74] Rathnayaka, S. D., C. Revoredo-Giha, and B. de Roos (2024) ““Assessing Scotland’s self-sufficiency of major food commodities””. In: *Agriculture & Food Security* 13.1 (). DOI: [10.1186/s40066-024-00486-0](https://doi.org/10.1186/s40066-024-00486-0). URL: <https://pure.sruc.ac.uk/en/publications/assessing-scotlands-self-sufficiency-of-major-food-commodities/> (visited on 09/19/2026) (cited on page 30).
- [75] Regjeringen (2026) “*Jordbruksavtalen styrker norsk matproduksjon i en urolig verden*”. Regjeringen.no. URL: <https://www.regjeringen.no/no/aktuelt/jordbruksavtalen-styrker-norsk-matproduksjon-i-en-urolig-verden/id3160129/> (cited on page 30).

- [76] Regjeringen (2024) “*Milepæl for norsk beredskap*”. Regjeringen.no. URL: <https://www.regjeringen.no/no/aktuelt/milepal-for-norsk-beredskap/id3046671/> (cited on page 31).
- [77] Robert Gordon University (2021) “*Majority of UK offshore workforce to be delivering low carbon energy By 2030*”. URL: <https://www.rgu.ac.uk/news/news-2021/majority-of-uk-offshore-workforce-to-be-delivering-low-carbon-energy-by-2030> (cited on page 8).
- [78] Sayers, P. et al. (2021) “*Third UK Climate Change Risk Assessment Technical Report*”. Tech. rep. UK Climate Risk. URL: <https://greenfutures.exeter.ac.uk/wp-content/uploads/2022/01/Technical-Report-The-Third-Climate-Change-Risk-Assessment.pdf> (cited on page 23).
- [79] Scottish Government (2024) “*Agriculture and Rural Communities (Scotland) Act 2024*”. URL: <https://www.legislation.gov.uk/asp/2024/11> (cited on page 33).
- [80] Scottish Government (2025) “*Biodiversity: planning guidance*”. URL: <https://www.gov.scot/publications/biodiversity-planning-guidance/> (cited on page 24).
- [81] Scottish Government (2024) “*Building a New Scotland*”. www.gov.scot. URL: <https://www.gov.scot/newscotland/> (visited on 09/19/2026) (cited on page 38).
- [82] Scottish Government (2024) “*Green industrial strategy*”. Tech. rep. URL: <https://www.gov.scot/binaries/content/documents/govscot/publications/strategy-plan/2024/09/green-industrial-strategy/documents/green-industrial-strategy/green-industrial-strategy/govscot%3Adocument/green-industrial-strategy.pdf> (cited on page 18).
- [83] Scottish Government (2022) “*Hydrogen action plan*”. Tech. rep. URL: <https://www.gov.scot/binaries/content/documents/govscot/publications/strategy-plan/2022/12/hydrogen-action-plan/documents/hydrogen-action-plan/hydrogen-action-plan/govscot%3Adocument/hydrogen-action-plan.pdf> (cited on page 17).
- [84] Scottish Government (2026) “*Legislative Consent Memorandum: Cyber Security and Resilience (Network and Information Systems) Bill*”. URL: <https://www.parlamaid-alba.scot/-/media/files/legislation/bills/lcms/cyber-security-and-resilience-network-and-information-systems-bill/splcms670.pdf> (cited on page 27).
- [85] Scottish Government (2026) “*National Islands Plan*”. Scottish Government. URL: <https://www.gov.scot/binaries/content/documents/govscot/publications/strategy-plan/2026/02/national-islands-plan-2/documents/national-islands-plan/national-islands-plan/govscot%3Adocument/national-islands-plan.pdf> (visited on 09/19/2026) (cited on page 38).

- [86] Scottish Government (2023) “*National Planning Framework 4*”. URL: <https://www.gov.scot/publications/national-planning-framework-4/> (cited on page 24).
- [87] Scottish Government (2023) “*Our marine sector in an independent Scotland*”. Tech. rep. URL: <https://www.gov.scot/binaries/content/documents/govscot/publications/research-and-analysis/2023/11/building-new-scotland-marine-sector-independent-scotland/documents/marine-sector-independent-scotland-paper-8/marine-sector-independent-scotland-paper-8/govscot%3Adocument/marine-sector-independent-scotland-paper-8.pdf> (visited on 09/19/2026) (cited on page 39).
- [88] Scottish Government (2026) “*Poverty and income inequality in Scotland 2022-25*”. Tech. rep. Scottish Government. URL: <https://www.gov.scot/publications/poverty-income-inequality-scotland-2022-25/> (cited on page 29).
- [89] Scottish Government (2025) “*Record levels of peatland restored*”. URL: <https://www.gov.scot/news/record-levels-of-peatland-restored/> (cited on page 32).
- [90] Scottish Government (2015) “*Scotland’s National Marine Plan: A Single Framework for Managing Our Seas*”. Chapter 14: Submarine Cables, policy CABLES 2. URL: <https://www.gov.scot/publications/scotlands-national-marine-plan/> (cited on page 24).
- [91] Scottish Government (2023) “*Scottish Greenhouse Gas Statistics 2023*”. Tech. rep. URL: <https://www.gov.scot/binaries/content/documents/govscot/publications/statistics/2025/06/scottish-greenhouse-gas-statistics-2023/documents/scottish-greenhouse-gas-statistics-2023/scottish-greenhouse-gas-statistics-2023/govscot%3Adocument/OFFICIAL%2BSENSITIVE%2B-%2Bpre%2Brelease%2Bstatistics%2B-%2B2023%2BGHG%2BPublication3.pdf> (cited on page 17).
- [92] Scottish Government (2016) “*The impact of diversity of ownership scale on social, economic and environmental outcomes*.” Tech. rep. URL: <https://www.gov.scot/binaries/content/documents/govscot/publications/research-and-analysis/2016/07/impact-diversity-ownership-scale-social-economic-environmental-outcomes/documents/00502355-pdf/00502355-pdf/govscot%3Adocument/00502355.pdf> (visited on 09/19/2026) (cited on page 33).
- [93] Scottish Government (2024) “*The Rural Support (Improvement) (Miscellaneous Amendment) (Scotland) Regulations 2024: Policy Note*”. URL: https://www.legislation.gov.uk/ssi/2024/380/pdfs/ssipn_20240380_en_001.pdf (cited on page 31).
- [94] Scottish Government (2025) “*The Scottish Health Survey 2024 - volume 1: main report*”. URL: <https://www.gov.scot/publications/scottish-health-survey-2024-volume-1-main-report/> (cited on page 29).
- [95] Scottish Land Commission (2025) “*Land Reform (Scotland) Act 2025*”. URL: <https://www.landcommission.gov.scot/shaping-land-policy/land-reform-act> (cited on page 33).

- [96] Scottish Renewables (2023) “*Onshore wind sector deal for Scotland*”. Tech. rep. Scottish Government. URL: <https://www.gov.scot/binaries/content/documents/govscot/publications/strategy-plan/2023/09/onshore-wind-sector-deal-scotland/documents/onshore-wind-sector-deal-scotland/onshore-wind-sector-deal-scotland/govscot%3Adocument/onshore-wind-sector-deal-scotland.pdf> (cited on page 17).
- [97] Scottish Water (2026) “*Our vital role*”. URL: <https://www.scottishwater.co.uk/About-Us/Who-We-Are/Our-Vital-Role> (cited on page 23).
- [98] Scottish Water (2026) “*Strategic review of charges 2027-33*”. Tech. rep. URL: <https://www.scottishwater.co.uk/-/media/scottishwater/document-hub/key-publications/delivery-and-business-plans/sw-responses-to-wics-draft-determination/scottish-waters-response-to-wics-draft-determination-2027-33.pdf> (cited on page 23).
- [99] ScottishPower (2025) “*ScottishPower to invest up to a record £12bn in Scotland’s electricity grid*”. URL: https://www.scottishpower.com/w/scottishpower_to_invest_up_to_a_record_12bn_in_scotlands_electricity_grid (cited on page 25).
- [100] SP Energy Networks (2026) “*Our RIIO-T3 Business Plan*”. URL: https://www.spenergynetworks.co.uk/pages/riio_t3_business_plan.aspx (cited on page 25).
- [101] SP Energy Networks (2024) “*Western HVDC link*”. URL: https://www.spenergynetworks.co.uk/pages/western_hvdc_link.aspx (cited on page 22).
- [102] State Council of the People’s Republic of China (2021) “*Regulation on Protecting the Security of Critical Information Infrastructure*”. State Council Order No. 745. URL: https://www.gov.cn/zhengce/content/2021-08/17/content_5631671.htm (cited on page 27).
- [103] State Grid Corporation of China (2025) “*Stronger, Smarter, Greener: State Grid Powers Chinese Modernization*”. URL: http://en.sasac.gov.cn/2025/12/17/c_20201.htm (cited on page 27).
- [104] Statistisk sentralbyrå (2026) “*Land use and land cover*”. SSB. URL: <https://www.ssb.no/en/natur-og-miljo/areal/statistikk/arealbruk-og-arealressurser> (visited on 09/19/2026) (cited on page 32).
- [105] Statnett (2025) “*Interconnectors*”. URL: <https://www.statnett.no/en/The-power-system/interconnectors/> (cited on page 23).
- [106] Statnett (2025) “*Introduction to our strategy*”. URL: <https://www.statnett.no/en/about-statnett/our-strategy-electrification-for-a-new-era/introduction> (cited on page 27).

- [107] Stern, M. and J. Öjendal (2010) “Mapping the Security–Development Nexus: Conflict, Complexity, Cacophony, Convergence?” In: *Security Dialogue* 41.1 (), pp. 5–29. URL: <https://www.jstor.org/stable/26301183> (cited on pages 3, 6).
- [108] Sunday Post (2018) “Argyll traders call for solution to A83 closures that cost up to £60,000 a day”. [sundaypost.com](https://www.sundaypost.com/fp/we-cant-rest-and-we-have-nothing-to-be-thankful-for-argyll-traders-call-for-solution-to-a83-closures-that-cost-up-to-60000-a-day/). URL: <https://www.sundaypost.com/fp/we-cant-rest-and-we-have-nothing-to-be-thankful-for-argyll-traders-call-for-solution-to-a83-closures-that-cost-up-to-60000-a-day/> (cited on page 21).
- [109] The Scottish Government (2025) “Energy Statistics for Scotland - Q3 2025”. Tech. rep. Gov.scot. URL: <https://www.gov.scot/publications/energy-statistics-for-scotland-q3-2025/pages/gas-consumption/> (cited on pages 3, 17).
- [110] The Scottish Government (2025) “Scottish fish farm production survey 2024”. Gov.scot. URL: <https://www.gov.scot/publications/scottish-fish-farm-production-survey-2024/pages/5/> (visited on 09/19/2026) (cited on page 30).
- [111] Thompson, Z., E. Iordanidou, and A. Scrimgeour (2026) “Scottish House condition survey 2024: Key findings”. Tech. rep. Scottish Government. URL: <https://www.gov.scot/binaries/content/documents/govscot/publications/statistics/2026/02/scottish-house-condition-survey-2024-key-findings/documents/scottish-house-condition-survey-2024---key-findings/scottish-house-condition-survey-2024---key-findings/govscot%3Adocument/Key%2BFindings%2B2024%2B-%2BComplete%2Breport.pdf> (cited on page 16).
- [112] Transition Economics (2024) “Scotwind: The investment needed to secure manufacturing jobs in Scotland”. Tech. rep. URL: <https://www.stuc.org.uk/resources/scotwind-report-2024.pdf> (visited on 09/19/2026) (cited on page 18).
- [113] Transport Scotland (2023) “Access to Argyll and Bute (A83)”. Tech. rep. Gov.scot. URL: <https://www.transport.gov.scot/projects/access-to-argyll-and-bute-a83/project-details/> (cited on page 21).
- [114] Transport Scotland (2023) “Options assessment report - January 2023 - A83 Access to Argyll and Bute - Introduction”. Tech. rep. URL: <https://www.transport.gov.scot/publication/medium-term-strategy-options-assessment-report-january-2023-a83-access-to-argyll-and-bute/introduction/> (cited on page 21).
- [115] Transport Scotland (2017) “Record of Determination”. Tech. rep. Gov.scot. URL: <https://www.transport.gov.scot/publication/draft-orders-medium-term-solution-a83-rest-and-be-thankful/record-of-determination/> (cited on page 21).
- [116] Trussell (2024) “State of Hunger 2024”. Tech. rep. Trussell. URL: <https://www.trussell.org.uk/> (cited on page 29).

- [117] UK Government (2026) “*Cyber Security and Resilience (Network and Information Systems) Bill: Explanatory Notes*”. URL: <https://publications.parliament.uk/pa/bills/lbill/59-02/032/en/5902032en06.htm> (cited on page 27).
- [118] UK Government (2025) “*Cyber Security and Resilience (Network and Information Systems) Bill: Summary*”. Proposed legislation; not yet enacted. URL: <https://www.gov.uk/government/publications/cyber-security-and-resilience-network-and-information-systems-bill-factsheets/summary-of-the-bill> (cited on page 25).
- [119] UK Government (1998) “*Scotland Act 1998, Schedule 5: Reserved Matters*”. URL: <https://www.legislation.gov.uk/ukpga/1998/46/schedule/5> (cited on page 26).
- [120] UK Government (2026) “*Written questions and answers: COBR Directorate responsibilities*”. URL: <https://questions-statements.parliament.uk/written-questions/detail/2026-01-06/103377/> (cited on page 26).
- [121] UK Tech News (2026) ““Over three-quarters of CNI organisations report repeated supply chain compromise””. In: *UK Tech News* (). URL: <https://uktechnews.co.uk/2026/07/30/over-three-quarters-of-cni-organisations-report-repeated-supply-chain-compromise-e2e-assure-research-reveals/> (cited on page 25).
- [122] UNDP (1994) “*Human development report 1994*”. Tech. rep. United Nations Development Programme. URL: <https://hdr.undp.org/system/files/documents/hdr1994encompletenostats.pdf> (cited on page 6).
- [123] UNDP (2022) “*New threats to human security in the anthropocene demanding greater solidarity*”. Tech. rep. United Nations Development Programme. URL: <https://hdr.undp.org/system/files/documents/srhs2022.pdf> (cited on page 6).
- [124] Venturi, B. (2017) “*The Security-Migration-Development Nexus revised*”. Tech. rep. Foundation for European Progressive Studies & Istituto Affari Internazionali. URL: <https://feps-europe.eu/wp-content/uploads/2018/02/The-security-migration-development-nexus-revised-a-perspective-from-the-Sahel.pdf> (cited on page 6).
- [125] World Ports Organization (2026) ““Green light from Ofgem for 2 GW subsea electricity superhighway””. In: *World Ports Organization* (). URL: <https://www.worldports.org/green-light-from-ofgem-for-2-gw-subsea-electricity-superhighway/> (cited on page 25).
- [126] Zhu, X. et al. (2024) ““The contribution of natural and anthropogenic causes to soil acidification rates under different fertilization practices and site conditions in southern China””. In: *Science of The Total Environment* 934.172986 (). DOI: [10.1016/j.scitotenv.2024.172986](https://doi.org/10.1016/j.scitotenv.2024.172986). URL: <https://www.sciencedirect.com/science/article/pii/S0048969724031334> (visited on 09/19/2026) (cited on page 32).



Sustainable Scotland:

A Comparative Analysis of Development-Security

21st September 2026

ID: 2401049

Name: Joshua McManus

Institution: University of Buckingham

School: Faculty of Business, Humanities, and Social Sciences

BA(Hons) Security, Intelligence, and Cyber